TEAM ASSEMBLY HANDOVER

Autonomous garbage collection rover | E2 baseline d6cd479 | 8 November 2026

START NOW: checkpoint A - disconnected mounting, reference labels, proposed layout and route measurements. Protected detached signal assembly follows verified component/package mapping.

KEEP DISCONNECTED: battery main/balance, every supply/USB during assembly, actual motor loads, unresolved protection/power ports and actual ECHO-to-MCU plugs. F-J energized stages are NOT RELEASED.

201 E2 records = 197 external segments + 4 internal-bond inspections (NO added wires). E2 baseline has 119 passing software tests; these do not establish actual compatibility. No supplied hardware photographs; no actual terminal matches yet.

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Remote review: send labeled top/underside and terminal photographs, exact-part documents, completed wire/terminal/measurement records and a checkpoint evidence batch. PENDING is never PASS.

1. Actual component identification

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

No hardware photographs were supplied in docs/hardware_photos/ at generation. Actual labels and terminal orientation are NOT IDENTIFIED. The following component and port names come from E2, not from observed hardware.

Assign the E2 reference to each physical item with a removable label. Photograph the full item and a readable terminal close-up before terminating a harness. Never infer header order from the picture of another board.

The battery stays unplugged, outside electrical work. Photograph its intact exterior and connector housings only. Do not unwrap the pack, open it or probe its balance pins for identification.

Use the photo request list below. Send original-resolution images with reference names and a ruler where mounting/connector dimensions matter. Add manufacturer datasheets and a terminal-map worksheet; photographs establish only what is visibly marked.

Current photograph inventory: NOT SUPPLIED (0 files).

E2 references	Inventory role	Actual identification status
ESP1	ESP32 DevKit	GPIO/power/header and USB path; actual board unknown
DRV_L / DRV_R	Two BTS modules	All twelve functional ports per module; orientation/current unconfirmed
DRV_A	One conditional L298 carrier	Regulator/enable links, supply labels, diode paths and thermal capability unconfirmed
US_F/L/R	Three ultrasonic modules	Actual four-pin header labels unobserved
BUCK1/2	Two LM2596 modules	Input/output polarity/bonds and continuous capacity unconfirmed
M1-M7	Seven motors	Individual identities/lead labels and current profiles unconfirmed
BT1	Battery	User inventory: nominal 11.1V/3300mAh; actual label/chemistry unknown
P1, OV1/2, K1, S0/2/3	Required protection/stop assemblies	E2 functional names only; actual parts and terminal maps not selected
U1/U2 and passives	Fixed E2 interface selections	Selected part codes/values, not physically observed

Exact identification photographs needed

Request / references	Requested filenames	Visible evidence and E2 comparison
P00 / whole rover	P00_rover_top.jpg; P00_rover_underplate.jpg; P00_plate_with_ruler.jpg	Front arrow, all mounts, separate cable routes, guarded brush/conveyor clearance, electronics plate dimensions; photograph battery detached. E2 match: Physical layout only; no hidden wire or terminal inference.
P01 / ESP1	P01_ESP1_front.jpg; P01_ESP1_back.jpg; P01_ESP1_headers_usb.jpg	Readable module can/chip/board revision, GPIO and power labels on both headers, EN/reset and USB connector type. E2 match: Match GPIO16/17/18/19/21/22/23/25/14/27/32/34/33/35/13/36, 3V3/GND and board-approved 5V input. USB power path needs schematic, not just photos.
P02 / DRV_L	P02_DRV_L_front.jpg; P02_DRV_L_back.jpg; P02_DRV_L_terminals.jpg	Chip/buffer markings, control header order and all screw labels, jumpers/links, connectors and heatsink mounting. Do not dismantle powered equipment. E2 match: B+/B-/M+/M-/VCC/GND/RPWM/LPWM/R_EN/L_EN/R_IS/L_IS, orientation and link configuration; no continuous-current inference.
P03 / DRV_R	P03_DRV_R_front.jpg; P03_DRV_R_back.jpg; P03_DRV_R_terminals.jpg	Same independent views as DRV_L; include this board's own labels and revision. E2 match: Match this module separately; identify internal B-/GND bond by later unpowered method.
P04 / DRV_A	P04_DRV_A_front.jpg; P04_DRV_A_back.jpg; P04_DRV_A_jumpers_diodes.jpg	Chip/regulator markings, Vs/5V/GND, OUT1-4, IN1-4, ENA/ENB labels, separate regulator jumper and all recirculation diodes. E2 match: Map actual carrier labels to E2 Vs_12V and 5V_LOGIC functional ports; schematic and thermal/load data still required.
P05 / US_F / US_L / US_R	P05_US_F_front_back.jpg; P05_US_L_front_back.jpg; P05_US_R_front_back.jpg	Each sensor face/rear and readable VCC/TRIG/ECHO/GND labels, version and mounting direction; paired views may share a photo if legible. E2 match: Three individual header maps; visual appearance does not prove ECHO levels.
P06 / BUCK1 / BUCK2	P06_BUCK1_front_back.jpg; P06_BUCK2_front_back.jpg; P06_buck_terminals.jpg	Each actual IN+/IN-/OUT+/OUT-, chip, inductor, diode, capacitor voltage markings and adjustment screw. E2 match: MCU versus logic designation and polarity; internal negative bond and capability need measurement/data.
P07 / M1-M7	P07_M1_label_leads.jpg through P07_M7_label_leads.jpg; P07_motor_mounts.jpg	Each distinct label/part number, rated voltage/speed if printed, gearbox, pigtails/connectors and mechanical direction arrow. Conveyor is a priority. E2 match: Assign motor identities and lead labels A/B provisionally. No operating/start/stall current or direction inferred from color/RPM.
P08 / BT1	P08_BT1_label.jpg; P08_BT1_connectors.jpg	Intact exterior chemistry/cell/capacity/C-rating/charge information and manufacturer; unplugged main/balance housings without exposing or probing contacts. E2 match: Confirm actual pack specification. Balance pin order must follow manufacturer data; do not unwrap or open battery.
P09 / U1 / U2 and interface board	P09_U1_mark_pin1.jpg; P09_U2_carrier_both_sides.jpg; P09_interface_top.jpg; P09_interface_underside.jpg	IC order code and pin1, actual carrier model/header markings, resistor reference/value labels, divider nodes, bypass placement and solder joints. E2 match: Verify AHCT244N and LVC3G34DCU, IC-to-carrier mapping and MCU3V3 supply for U2; attach header continuity map.
P10 / P1 / OV1 / OV2 / K1 / S0 / S2 / S3	P10_<reference>_label_terminals.jpg for each actual selected assembly	Manufacturer/part/revision, complete terminal legends, coil/contact blocks and markings; attach maker datasheet and a function-to-terminal mapping. E2 match: These assemblies are unselected E2 functional ports; photographs alone do not close ratings, thresholds, isolated contact or reset behavior.
P11 / F0 / F1 / F2 / FK / FL / FR / FA / FM1-7 / TB_POS / TB_GND / clamps / capacitors / harness	P11_protection_parts_labels.jpg; P11_distribution_terminals.jpg; P11_harness_markings.jpg	Actual fuse/holder series and rating/curve references, wire/connector markings, capacitor/clamp codes and polarity, mounting and conductor-length schedule. E2 match: Match approved selections only after current/thermal/fault criteria exist; unmarked parts remain unidentified.
P12 / checkpoint evidence	P12_<checkpoint>_<view_or_wire_ID>.jpg	Unpowered overview, each terminal strip close-up, both ends of labeled harnesses; probe endpoints and instrument display for ambiguous/changed routes. Provide a reading row for every connection, not 201 repetitive photos. E2 match: Tie every measurement and photo to wire IDs/checkpoint; preserve originals and explain any deviation.

If images are later added, they require visual review and a recorded observation before any terminal can be marked matched. Photo filenames alone do not close a gate.

2. Rover top-down layout

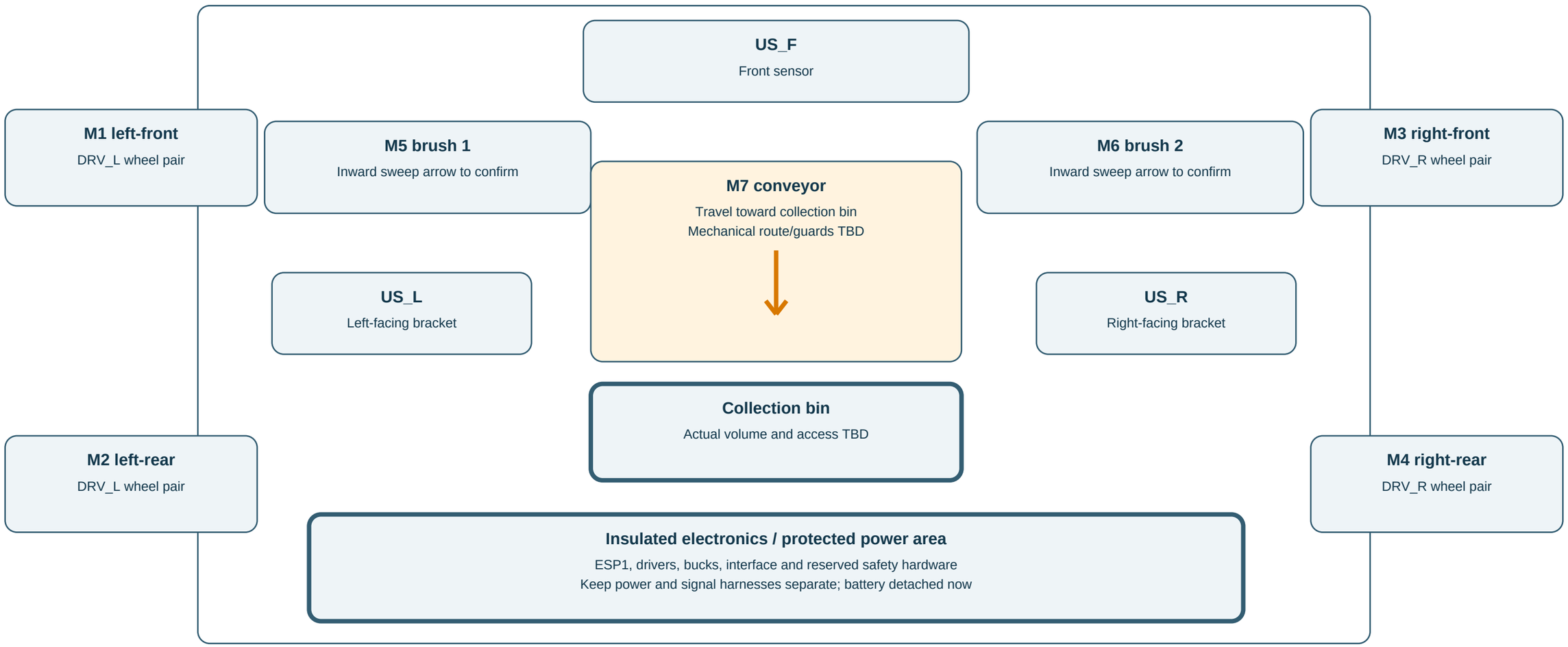
Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

The top-down drawing is a proposed placement plan, not a measured mechanical drawing. FRONT is the sensor/cleaning end. M1/M2 are the left wheel pair, M3/M4 the right pair; M5/M6 brushes and M7 conveyor remain distinct labels.

Dry-fit the chassis, motor mounts, sensor brackets, guarded cleaning head, collection bin and insulated electronics plate. Verify real clearances and photograph the layout before drilling final positions. Do not cut or modify motor leads.

Reserve separate routes for high-current motor/power conductors and logic/sensor harnesses. Mark intended cable routes with string and measure their lengths; leave P1/OV1/OV2/K1/fuse/distribution mounting areas provisional until the actual selected parts fit.

The E2 physical-harness diagram follows the proposed placement drawing. Equal labels describe electrical nets; bundled lines do not establish connector pin order or exact cable lengths.

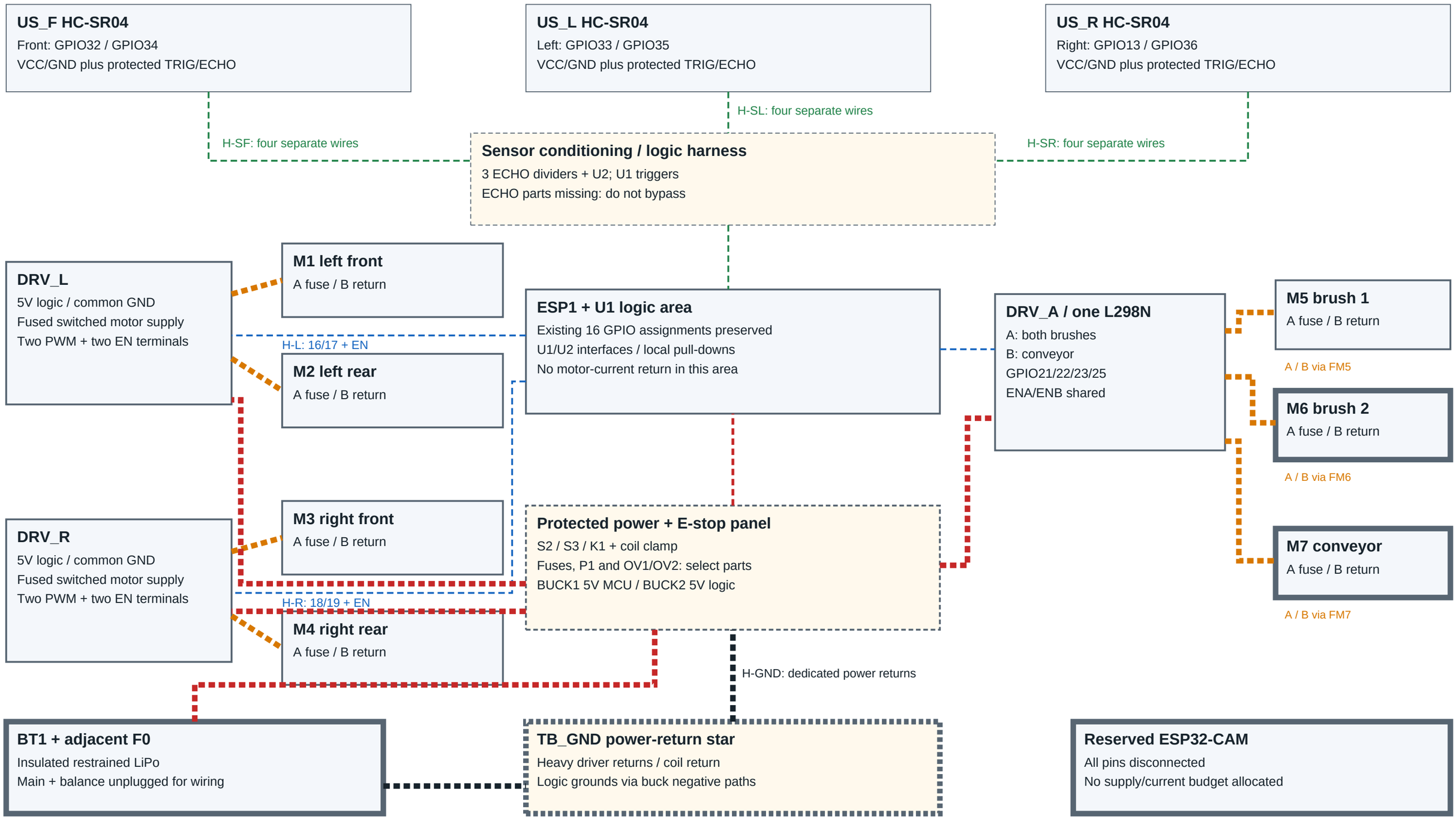


Sensor/brush/bin placements are proposals. Remote supervisor accepts actual layout from checkpoint A photos.

Physical harness layout - placement suggestion

Not to scale or connector orientation. Bundles are separate labeled conductors; use the full terminal/connection tables before fitting any wire.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026



Keep high-current orange/red/black wiring separate from blue/green signal harnesses. Use insulated rated terminals; no breadboard motor paths. Exact wire gauge, length, connectors, routing and suppression remain release items.

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

All seven motor terminals

Equal net names join. Exact connections and protection: assembly_connections sheets / READY_TO_WIRE.md. Ratings remain HOLD.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

M1 | left front wheel motor
INVENTORY / VERIFY
12 V / 30 RPM

A ---- M1_A
M
B ---- DRV_L_MN

M2 | left rear wheel motor
INVENTORY / VERIFY
12 V / 30 RPM

A ---- M2_A
M
B ---- DRV_L_MN

M3 | right front wheel motor
INVENTORY / VERIFY
12 V / 30 RPM

A ---- M3_A
M
B ---- DRV_R_MN

M4 | right rear wheel motor
INVENTORY / VERIFY
12 V / 30 RPM

A ---- M4_A
M
B ---- DRV_R_MN

M5 | brush 1 motor
INVENTORY / VERIFY
12 V

A ---- M5_A
M
B ---- DRV_A_OUT2

M6 | brush 2 motor
INVENTORY / VERIFY
12 V

A ---- M6_A
M
B ---- DRV_A_OUT2

M7 | conveyor motor
INVENTORY / VERIFY
Voltage TBD

A ---- M7_A
M
B ---- DRV_A_OUT4

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

3. ESP32 connections

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

Use GPIO numbers, not a guessed DevKit header position. Confirm ESP32-WROOM-32 and the actual board markings. Record 5V/VIN, 3V3, GND, EN/reset, USB connector and all used GPIO header positions in the terminal-map worksheet.

Keep USB, external supplies and battery disconnected during assembly. A detached signal-interface harness may be prepared after its endpoints and conductor requirements are established. Do not connect MCU power, unknown feedback contacts or real ECHO until their gates close.

E2 GPIOs are unchanged. U1 handles wheel PWM, shared enable and TRIG. GPIO21/22 and 23/25 serve the conditional L298 inputs. All ECHO inputs go through U2; GPIO27 is HIGH healthy through isolated contacts with a pull-down.

For USB-only installation later, disconnect every external power and GPIO harness from the DevKit. Firmware starts disabled; open GPIO27 intentionally latches ESTOP. Do not bridge the feedback HIGH to get past it.

Firmware signal	GPIO	E2 endpoint / net
L_FWD	16	ESP1.GPIO16 / L_FWD_3V3
L_REV	17	ESP1.GPIO17 / L_REV_3V3
R_FWD	18	ESP1.GPIO18 / R_FWD_3V3
R_REV	19	ESP1.GPIO19 / R_REV_3V3
ENABLE	14	ESP1.GPIO14 / ENABLE_3V3
TF	32	ESP1.GPIO32 / TF_3V3
TL	33	ESP1.GPIO33 / TL_3V3
TR	13	ESP1.GPIO13 / TR_3V3
BRUSH_IN1	21	ESP1.GPIO21 / BRUSH_IN1
BRUSH_IN2	22	ESP1.GPIO22 / BRUSH_IN2
CONVEYOR_IN3	23	ESP1.GPIO23 / CONVEYOR_IN3
CONVEYOR_IN4	25	ESP1.GPIO25 / CONVEYOR_IN4
EF	34	ESP1.GPIO34 / EF_SAFE
EL	35	ESP1.GPIO35 / EL_SAFE
ER	36	ESP1.GPIO36 / ER_SAFE
ESTOP_OK	27	ESP1.GPIO27 / ESTOP_OK

ESP32 pinout connection chart

Functional GPIO labels only. Exact DevKit header positions/USB power path remain unverified; no wiring orientation implied.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

GPIO / terminal	Firmware signal	Exact proposed endpoint chain	Electrical status / requirement
GPIO16	L_FWD	U1.2 -> U1.18 -> DRV_L.RPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO17	L_REV	U1.4 -> U1.16 -> DRV_L.LPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO18	R_FWD	U1.6 -> U1.14 -> DRV_R.RPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO19	R_REV	U1.8 -> U1.12 -> DRV_R.LPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO14	ENABLE	U1.11 -> U1.9 -> S2 EN NC -> six enables	**BLOCKED_MISSING** U1 + bias + cutoff missing
GPIO32	TF	U1.13 -> U1.7 -> US_F.TRIG	**BLOCKED_MISSING** U1 + bias parts missing
GPIO33	TL	U1.15 -> U1.5 -> US_L.TRIG	**BLOCKED_MISSING** U1 + bias parts missing
GPIO13	TR	U1.17 -> U1.3 -> US_R.TRIG	**BLOCKED_MISSING** U1 + bias parts missing
GPIO21	BRUSH_IN1	DRV_A.IN1 / both brushes	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO22	BRUSH_IN2	DRV_A.IN2 / held LOW	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO23	CONVEYOR_IN3	DRV_A.IN3 / conveyor	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO25	CONVEYOR_IN4	DRV_A.IN4 / held LOW	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO34	EF	ECHO -> R23/R24 -> U2.1/7 -> GPIO34	**BLOCKED_MISSING** ECHO disconnected; parts missing
GPIO35	EL	ECHO -> R25/R26 -> U2.3/5 -> GPIO35	**BLOCKED_MISSING** ECHO disconnected; parts missing
GPIO36	ER	ECHO -> R27/R28 -> U2.6/2 -> GPIO36 / VP	**BLOCKED_MISSING** ECHO disconnected; parts missing
GPIO27	ESTOP_OK	3V3 -> S2 NC -> K1 sense NO -> GPIO27	**BLOCKED_MISSING** Cutoff/feedback parts missing

5V/VIN: only V5_MCU after OV1 and actual-board qualification. 3V3: E-stop dry feedback + U2 only. GND: logic return to BUCK1 OUT-.
Never feed 5V into a GPIO or 3V3 pin.

Unused: GPIO26, GPIO39/VN, GPIO4. Reserve GPIO1/3 for serial, GPIO0/2/5/12/15 boot straps, GPIO6-11 flash. Keep EN/reset under onboard control; no JTAG on GPIO13/14. ESP32-CAM remains disconnected.

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked
Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Detailed connection records - ESP32 connections

A record is one E2 segment. Endpoint resistor/fuse/contact bodies continue into other records; do not install duplicate components or jumpers. Color identifies function; voltage and ratings are requirements, not measurements.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W023 | EXTERNAL CONNECTION - gate dependent **Source:** OV1.OUT+ **Destination:** ESP1.5V_VIN **Identification:** Red; label both ends W023; V5_MCU **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV1 [5V rail 1 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Measure output unloaded/loaded and verify actual DevKit regulated-5V input and USB backfeed path. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W036 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.3V3 **Destination:** U2.8 (VCC) **Identification:** Red; label both ends W036; V3V3 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: MCU-derived 3V3; dry feedback and U2 only; never V5_LOGIC.	**Required electrical rating:** ESP32 3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify no connection to 5V/coil circuit and actual feedback-contact minimum current. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W037 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.3V3 **Destination:** C_U2.1 **Identification:** Red; label both ends W037; V3V3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_U2 [local ceramic bypass; 100nF, >=10V]. Surrounding protection: MCU-derived 3V3; dry feedback and U2 only; never V5_LOGIC.	**Required electrical rating:** ESP32 3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify no connection to 5V/coil circuit and actual feedback-contact minimum current. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W040 | EXTERNAL CONNECTION - gate dependent **Source:** BUCK1.OUT- **Destination:** ESP1.GND **Identification:** Black; label both ends W040; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W058 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GND **Destination:** R24.2 **Identification:** Black; label both ends W058; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R24 [ECHO lower resistor; 3.3k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W059 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GND **Destination:** R26.2 **Identification:** Black; label both ends W059; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R26 [ECHO lower resistor; 3.3k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W060 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GND **Destination:** R28.2 **Identification:** Black; label both ends W060; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R28 [ECHO lower resistor; 3.3k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W083 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GND **Destination:** R29.2 **Identification:** Black; label both ends W083; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R29 [E-stop feedback pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W096 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GND **Destination:** U2.4 (GND) **Identification:** Black; label both ends W096; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W098 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GND **Destination:** R30.2 **Identification:** Black; label both ends W098; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R30 [ECHO output default-low; 100k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W099 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GND **Destination:** R31.2 **Identification:** Black; label both ends W099; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R31 [ECHO output default-low; 100k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W100 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GND **Destination:** R32.2 **Identification:** Black; label both ends W100; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R32 [ECHO output default-low; 100k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W113 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO16 **Destination:** U1.2 (1A1) **Identification:** Blue; label both ends W113; L_FWD_3V3 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: U1 TTL-compatible input; pull-down R1.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W114 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO16 **Destination:** R1.1 **Identification:** Blue; label both ends W114; L_FWD_3V3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R1 [L_FWD input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: U1 TTL-compatible input; pull-down R1.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W117 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO17 **Destination:** U1.4 (1A2) **Identification:** Blue; label both ends W117; L_REV_3V3 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: U1 TTL-compatible input; pull-down R2.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W118 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO17 **Destination:** R2.1 **Identification:** Blue; label both ends W118; L_REV_3V3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R2 [L_REV input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: U1 TTL-compatible input; pull-down R2.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W121 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO18 **Destination:** U1.6 (1A3) **Identification:** Blue; label both ends W121; R_FWD_3V3 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: U1 TTL-compatible input; pull-down R3.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W122 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO18 **Destination:** R3.1 **Identification:** Blue; label both ends W122; R_FWD_3V3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R3 [R_FWD input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: U1 TTL-compatible input; pull-down R3.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W125 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO19 **Destination:** U1.8 (1A4) **Identification:** Blue; label both ends W125; R_REV_3V3 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: U1 TTL-compatible input; pull-down R4.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W126 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO19 **Destination:** R4.1 **Identification:** Blue; label both ends W126; R_REV_3V3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R4 [R_REV input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: U1 TTL-compatible input; pull-down R4.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W129 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO14 **Destination:** U1.11 (2A1) **Identification:** Blue; label both ends W129; ENABLE_3V3 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: U1 TTL-compatible input; pull-down R5.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W130 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO14 **Destination:** R5.1 **Identification:** Blue; label both ends W130; ENABLE_3V3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R5 [ENABLE input pull-down; 2.2k 1%, >=1/8 W]. Surrounding protection: U1 TTL-compatible input; pull-down R5.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W132 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO32 **Destination:** U1.13 (2A2) **Identification:** Green; label both ends W132; TF_3V3 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: U1 TTL-compatible input; pull-down R6.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W133 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO32 **Destination:** R6.1 **Identification:** Green; label both ends W133; TF_3V3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R6 [TF input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: U1 TTL-compatible input; pull-down R6.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W135 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO33 **Destination:** U1.15 (2A3) **Identification:** Green; label both ends W135; TL_3V3 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: U1 TTL-compatible input; pull-down R7.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W136 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO33 **Destination:** R7.1 **Identification:** Green; label both ends W136; TL_3V3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R7 [TL input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: U1 TTL-compatible input; pull-down R7.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W138 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO13 **Destination:** U1.17 (2A4) **Identification:** Green; label both ends W138; TR_3V3 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: U1 TTL-compatible input; pull-down R8.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W139 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO13 **Destination:** R8.1 **Identification:** Green; label both ends W139; TR_3V3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R8 [TR input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: U1 TTL-compatible input; pull-down R8.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Continuity and scope at reset/boot/brownout; do not bypass buffer without exact-board qualification. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W153 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO21 **Destination:** DRV_A.IN1 **Identification:** Blue; label both ends W153; BRUSH_IN1 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Genuine L298 TTL input; local 10k pull-down.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W154 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO21 **Destination:** R9.1 **Identification:** Blue; label both ends W154; BRUSH_IN1 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R9 [BRUSH_IN1 pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Genuine L298 TTL input; local 10k pull-down.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W155 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO22 **Destination:** DRV_A.IN2 **Identification:** Blue; label both ends W155; BRUSH_IN2 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Genuine L298 TTL input; local 10k pull-down.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W156 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO22 **Destination:** R10.1 **Identification:** Blue; label both ends W156; BRUSH_IN2 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R10 [BRUSH_IN2 pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Genuine L298 TTL input; local 10k pull-down.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W157 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO23 **Destination:** DRV_A.IN3 **Identification:** Blue; label both ends W157; CONVEYOR_IN3 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Genuine L298 TTL input; local 10k pull-down.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W158 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO23 **Destination:** R11.1 **Identification:** Blue; label both ends W158; CONVEYOR_IN3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R11 [CONVEYOR_IN3 pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Genuine L298 TTL input; local 10k pull-down.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W159 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO25 **Destination:** DRV_A.IN4 **Identification:** Blue; label both ends W159; CONVEYOR_IN4 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Genuine L298 TTL input; local 10k pull-down.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W160 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.GPIO25 **Destination:** R12.1 **Identification:** Blue; label both ends W160; CONVEYOR_IN4 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R12 [CONVEYOR_IN4 pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Genuine L298 TTL input; local 10k pull-down.	**Required electrical rating:** 0/3.3 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify exact module input circuit, common GND and inputs LOW at reset; IN2/IN4 held LOW by firmware. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W164 | EXTERNAL CONNECTION - gate dependent **Source:** U2.7 (1Y) **Destination:** ESP1.GPIO34 **Identification:** Green; label both ends W164; EF_SAFE **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: U2 Ioff supplied by ESP1.3V3; 100k output bias.	**Required electrical rating:** 0 / MCU-derived 3.3V; U2 off -> high impedance; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W169 | EXTERNAL CONNECTION - gate dependent **Source:** U2.5 (2Y) **Destination:** ESP1.GPIO35 **Identification:** Green; label both ends W169; EL_SAFE **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: U2 Ioff supplied by ESP1.3V3; 100k output bias.	**Required electrical rating:** 0 / MCU-derived 3.3V; U2 off -> high impedance; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W174 | EXTERNAL CONNECTION - gate dependent **Source:** U2.2 (3Y) **Destination:** ESP1.GPIO36 **Identification:** Green; label both ends W174; ER_SAFE **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: U2 Ioff supplied by ESP1.3V3; 100k output bias.	**Required electrical rating:** 0 / MCU-derived 3.3V; U2 off -> high impedance; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

4. Left BTS7960 wiring

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

Label this actual module DRV_L. Match every visible label to B+, B-, M+, M-, VCC, GND, RPWM, LPWM, R_EN, L_EN, R_IS and L_IS; unresolved or differently named positions remain unconnected.

M1 left-front and M2 left-rear share M+/M- in parallel, each A lead through its own FM fuse. The motor lead sizes, fuses, B+ branch and B- return remain conditional on load and conductor data; label their ends now without terminating unknown-rated power paths.

RPWM uses U1.18 from GPIO16; LPWM uses U1.16 from GPIO17. Both enable inputs use the hardware-gated shared enable with local pulls. Do not retain hard-HIGH enable links or alter unidentified links by appearance alone.

W046 checks the module's existing B--to-GND bond; it is not a second ground wire. R_IS/L_IS remain separately insulated, unconnected. No current rating is inferred from the BTS7960 name or a module sales claim.

Left BTS7960 and wheel pair

Equal net names join. Exact connections and protection: assembly_connections sheets / READY_TO_WIRE.md. Ratings remain HOLD.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

DRV_L | BTS7960 / IBT-2 left

INVENTORY / VERIFY

module terminal symbol

B+	LEFT_SUPPLY
B-	GND
M+	DRV_L_MP
M-	DRV_L_MN
VCC	V5_LOGIC
GND	GND
RPWM	L_FWD_5V
LPWM	L_REV_5V
R_EN	EN_DRIVER_5V
L_EN	EN_DRIVER_5V
R_IS	NC - INSULATE
L_IS	NC - INSULATE

M1 | left front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A		M1_A
B	M	DRV_L_MN

M2 | left rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A		M2_A
B	M	DRV_L_MN

FM1 | motor 1 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_L_MP
OUT		M1_A

FM2 | motor 2 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_L_MP
OUT		M2_A

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Detailed connection records - Left BTS7960 wiring

A record is one E2 segment. Endpoint resistor/fuse/contact bodies continue into other records; do not install duplicate components or jumpers. Color identifies function; voltage and ratings are requirements, not measurements.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W012 | EXTERNAL CONNECTION - gate dependent **Source:** FL.OUT **Destination:** DRV_L.B+ **Identification:** Red; label both ends W012; LEFT_SUPPLY **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FL [left driver fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FL; motor branch overload coordination.	**Required electrical rating:** Pack voltage; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify sum of two left start/stall currents and actual IBT-2 thermal/connector capacity. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W013 | EXTERNAL CONNECTION - gate dependent **Source:** FL.OUT **Destination:** C_BL.+ **Identification:** Red; label both ends W013; LEFT_SUPPLY **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_BL [local motor-bus reservoir; C / ESR / voltage TBD]; FL [left driver fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FL; motor branch overload coordination.	**Required electrical rating:** Pack voltage; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify sum of two left start/stall currents and actual IBT-2 thermal/connector capacity. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W014 | EXTERNAL CONNECTION - gate dependent **Source:** FL.OUT **Destination:** Z_L.+ **Identification:** Red; label both ends W014; LEFT_SUPPLY **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FL [left driver fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FL; motor branch overload coordination.	**Required electrical rating:** Pack voltage; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify sum of two left start/stall currents and actual IBT-2 thermal/connector capacity. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W025 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.OUT+ **Destination:** DRV_L.VCC **Identification:** Red; label both ends W025; V5_LOGIC **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; qualify 4.75-5.25 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W045 | EXTERNAL CONNECTION - gate dependent **Source:** TB_GND.BUS **Destination:** DRV_L.B- **Identification:** Black; label both ends W045; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W073 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_L.GND **Destination:** R13.2 **Identification:** Black; label both ends W073; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R13 [DRV_L.R_EN local EN pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W074 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_L.GND **Destination:** R14.2 **Identification:** Black; label both ends W074; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R14 [DRV_L.L_EN local EN pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W079 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_L.GND **Destination:** R19.2 **Identification:** Black; label both ends W079; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R19 [L_FWD receiver pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W080 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_L.GND **Destination:** R20.2 **Identification:** Black; label both ends W080; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R20 [L_REV receiver pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W088 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_L.B- **Destination:** C_BL.- **Identification:** Black; label both ends W088; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_BL [local motor-bus reservoir; C / ESR / voltage TBD]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W093 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_L.B- **Destination:** Z_L.- **Identification:** Black; label both ends W093; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W115 | EXTERNAL CONNECTION - gate dependent **Source:** U1.18 (1Y1) **Destination:** DRV_L.RPWM **Identification:** Blue; label both ends W115; L_FWD_5V **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W119 | EXTERNAL CONNECTION - gate dependent **Source:** U1.16 (1Y2) **Destination:** DRV_L.LPWM **Identification:** Blue; label both ends W119; L_REV_5V **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W176 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_L.M+ **Destination:** FM1.IN **Identification:** Orange; label both ends W176; DRV_L_MP **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM1 [motor 1 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W177 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_L.M+ **Destination:** FM2.IN **Identification:** Orange; label both ends W177; DRV_L_MP **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM2 [motor 2 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W178 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_L.M- **Destination:** M1.B **Identification:** Orange; label both ends W178; DRV_L_MN **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W179 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_L.M- **Destination:** M2.B **Identification:** Orange; label both ends W179; DRV_L_MN **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W180 | EXTERNAL CONNECTION - gate dependent **Source:** FM1.OUT **Destination:** M1.A **Identification:** Orange; label both ends W180; M1_A **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM1 [motor 1 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FM1; value/time curve TBD.	**Required electrical rating:** Bridge output after fuse; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W181 | EXTERNAL CONNECTION - gate dependent **Source:** FM2.OUT **Destination:** M2.A **Identification:** Orange; label both ends W181; M2_A **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM2 [motor 2 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FM2; value/time curve TBD.	**Required electrical rating:** Bridge output after fuse; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.

5. Right BTS7960 wiring

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

Label this actual module DRV_R and identify its own markings; do not assume it is the same revision or header orientation as DRV_L.

M3 right-front and M4 right-rear share this module's M+/M- in parallel, each A lead through FM3/FM4. Final lead polarity is determined only during later guarded individual-motor tests, never from insulation color.

RPWM uses U1.14 from GPIO18; LPWM uses U1.12 from GPIO19. Both enables use the same physical cutoff path as the left driver, with separate local pulls.

W048 checks the existing B--to-GND bond; do not add a thin parallel return. Keep motor and power terminals isolated until the actual rating, protection and terminal maps have been accepted.

Right BTS7960 and wheel pair

Equal net names join. Exact connections and protection: assembly_connections sheets / READY_TO_WIRE.md. Ratings remain HOLD.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

DRV_R | BTS7960 / IBT-2 right

INVENTORY / VERIFY

module terminal symbol

B+	RIGHT_SUPPLY
B-	GND
M+	DRV_R_MP
M-	DRV_R_MN
VCC	V5_LOGIC
GND	GND
RPWM	R_FWD_5V
LPWM	R_REV_5V
R_EN	EN_DRIVER_5V
L_EN	EN_DRIVER_5V
R_IS	NC - INSULATE
L_IS	NC - INSULATE

M3 | right front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A		M3_A
B	M	DRV_R_MN

M4 | right rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A		M4_A
B	M	DRV_R_MN

FM3 | motor 3 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_R_MP
OUT		M3_A

FM4 | motor 4 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_R_MP
OUT		M4_A

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Detailed connection records - Right BTS7960 wiring

A record is one E2 segment. Endpoint resistor/fuse/contact bodies continue into other records; do not install duplicate components or jumpers. Color identifies function; voltage and ratings are requirements, not measurements.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W015 | EXTERNAL CONNECTION - gate dependent **Source:** FR.OUT **Destination:** DRV_R.B+ **Identification:** Red; label both ends W015; RIGHT_SUPPLY **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FR [right driver fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FR; motor branch overload coordination.	**Required electrical rating:** Pack voltage; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify sum of two right start/stall currents and actual IBT-2 thermal/connector capacity. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W016 | EXTERNAL CONNECTION - gate dependent **Source:** FR.OUT **Destination:** C_BR.+ **Identification:** Red; label both ends W016; RIGHT_SUPPLY **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_BR [local motor-bus reservoir; C / ESR / voltage TBD]; FR [right driver fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FR; motor branch overload coordination.	**Required electrical rating:** Pack voltage; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify sum of two right start/stall currents and actual IBT-2 thermal/connector capacity. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W017 | EXTERNAL CONNECTION - gate dependent **Source:** FR.OUT **Destination:** Z_R.+ **Identification:** Red; label both ends W017; RIGHT_SUPPLY **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FR [right driver fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FR; motor branch overload coordination.	**Required electrical rating:** Pack voltage; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify sum of two right start/stall currents and actual IBT-2 thermal/connector capacity. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W026 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.OUT+ **Destination:** DRV_R.VCC **Identification:** Red; label both ends W026; V5_LOGIC **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; qualify 4.75-5.25 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W047 | EXTERNAL CONNECTION - gate dependent **Source:** TB_GND.BUS **Destination:** DRV_R.B- **Identification:** Black; label both ends W047; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W075 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_R.GND **Destination:** R15.2 **Identification:** Black; label both ends W075; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R15 [DRV_R.R_EN local EN pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W076 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_R.GND **Destination:** R16.2 **Identification:** Black; label both ends W076; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R16 [DRV_R.L_EN local EN pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W081 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_R.GND **Destination:** R21.2 **Identification:** Black; label both ends W081; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R21 [R_FWD receiver pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W082 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_R.GND **Destination:** R22.2 **Identification:** Black; label both ends W082; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R22 [R_REV receiver pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W089 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_R.B- **Destination:** C_BR.- **Identification:** Black; label both ends W089; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_BR [local motor-bus reservoir; C / ESR / voltage TBD]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W094 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_R.B- **Destination:** Z_R.- **Identification:** Black; label both ends W094; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W123 | EXTERNAL CONNECTION - gate dependent **Source:** U1.14 (1Y3) **Destination:** DRV_R.RPWM **Identification:** Blue; label both ends W123; R_FWD_5V **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W127 | EXTERNAL CONNECTION - gate dependent **Source:** U1.12 (1Y4) **Destination:** DRV_R.LPWM **Identification:** Blue; label both ends W127; R_REV_5V **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W182 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_R.M+ **Destination:** FM3.IN **Identification:** Orange; label both ends W182; DRV_R_MP **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM3 [motor 3 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W183 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_R.M+ **Destination:** FM4.IN **Identification:** Orange; label both ends W183; DRV_R_MP **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM4 [motor 4 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W184 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_R.M- **Destination:** M3.B **Identification:** Orange; label both ends W184; DRV_R_MN **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W185 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_R.M- **Destination:** M4.B **Identification:** Orange; label both ends W185; DRV_R_MN **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W186 | EXTERNAL CONNECTION - gate dependent **Source:** FM3.OUT **Destination:** M3.A **Identification:** Orange; label both ends W186; M3_A **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM3 [motor 3 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FM3; value/time curve TBD.	**Required electrical rating:** Bridge output after fuse; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W187 | EXTERNAL CONNECTION - gate dependent **Source:** FM4.OUT **Destination:** M4.A **Identification:** Orange; label both ends W187; M4_A **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM4 [motor 4 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FM4; value/time curve TBD.	**Required electrical rating:** Bridge output after fuse; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.

6. L298N cleaning system

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

The active E2 proposal is one L298N: IN1/IN2 from GPIO21/22, IN3/IN4 from GPIO23/25, ENA/ENB from the gated shared enable. OUT1/OUT2 feed both brushes M5/M6, with individual FM5/FM6; OUT3/OUT4 feed M7 through FM7.

Identify Vs/+12V, external 5V logic terminal, ground, all control/output labels, separate regulator jumper and ENA/ENB jumpers, regulator part, recirculation diodes and chip marking. E2 Vs_12V/5V_LOGIC are functional port names, not verified printed labels.

Do not attach external 5V until the actual carrier schematic establishes how its onboard regulator must be disabled. Do not confuse that regulator link with the enable jumpers. Keep all cleaning motor power leads unplugged.

Both brushes' currents add on channel A; conveyor current and package heating occur simultaneously on B. Motor voltage/RPM does not establish suitability. If current, delivered voltage or thermal limits fail, follow the conditional replacement/revision instructions in section 12; do not simply reduce software duty.

Both brush directions must sweep inward and conveyor travel must lead into the bin. Mark mechanical arrows now; establish electrical A/B polarity later with approved current-limited tests. Cleaning commands operate only during actual forward travel in the existing firmware.

Conditional L298N and cleaning motors

Equal net names join. Exact connections and protection: assembly_connections sheets / READY_TO_WIRE.md. Ratings remain HOLD.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

DRV_A | ONE L298N: A brushes / B conveyor

INVENTORY / VERIFY

module terminal symbol

Vs_12V	AUX_SUPPLY
5V_LOGIC	V5_LOGIC
GND	GND
IN1	BRUSH_IN1
IN2	BRUSH_IN2
IN3	CONVEYOR_IN3
IN4	CONVEYOR_IN4
ENA	EN_DRIVER_5V
ENB	EN_DRIVER_5V
OUT1	DRV_A_OUT1
OUT2	DRV_A_OUT2
OUT3	DRV_A_OUT3
OUT4	DRV_A_OUT4

M5 | brush 1 motor

INVENTORY / VERIFY

12 V

A		M5_A
B	M	DRV_A_OUT2

M6 | brush 2 motor

INVENTORY / VERIFY

12 V

A		M6_A
B	M	DRV_A_OUT2

M7 | conveyor motor

INVENTORY / VERIFY

Voltage TBD

A		M7_A
B	M	DRV_A_OUT4

FM5 | motor 5 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_A_OUT1
OUT		M5_A

FM6 | motor 6 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_A_OUT1
OUT		M6_A

FM7 | motor 7 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN		DRV_A_OUT3
OUT		M7_A

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Detailed connection records - L298N cleaning system

A record is one E2 segment. Endpoint resistor/fuse/contact bodies continue into other records; do not install duplicate components or jumpers. Color identifies function; voltage and ratings are requirements, not measurements.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W018 | EXTERNAL CONNECTION - gate dependent **Source:** FA.OUT **Destination:** DRV_A.Vs_12V **Identification:** Red; label both ends W018; AUX_SUPPLY **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FA [auxiliary driver fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FA + FM5/FM6/FM7; package thermal review.	**Required electrical rating:** Pack voltage; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify combined brush current, conveyor current, per-channel and shared thermal constraints. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W019 | EXTERNAL CONNECTION - gate dependent **Source:** FA.OUT **Destination:** C_BA.+ **Identification:** Red; label both ends W019; AUX_SUPPLY **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_BA [local motor-bus reservoir; C / ESR / voltage TBD]; FA [auxiliary driver fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FA + FM5/FM6/FM7; package thermal review.	**Required electrical rating:** Pack voltage; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify combined brush current, conveyor current, per-channel and shared thermal constraints. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W020 | EXTERNAL CONNECTION - gate dependent **Source:** FA.OUT **Destination:** Z_A.+ **Identification:** Red; label both ends W020; AUX_SUPPLY **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FA [auxiliary driver fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FA + FM5/FM6/FM7; package thermal review.	**Required electrical rating:** Pack voltage; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify combined brush current, conveyor current, per-channel and shared thermal constraints. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W027 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.OUT+ **Destination:** DRV_A.5V_LOGIC **Identification:** Red; label both ends W027; V5_LOGIC **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; qualify 4.75-5.25 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W049 | EXTERNAL CONNECTION - gate dependent **Source:** TB_GND.BUS **Destination:** DRV_A.GND **Identification:** Black; label both ends W049; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W069 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.GND **Destination:** R9.2 **Identification:** Black; label both ends W069; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R9 [BRUSH_IN1 pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W070 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.GND **Destination:** R10.2 **Identification:** Black; label both ends W070; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R10 [BRUSH_IN2 pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W071 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.GND **Destination:** R11.2 **Identification:** Black; label both ends W071; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R11 [CONVEYOR_IN3 pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W072 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.GND **Destination:** R12.2 **Identification:** Black; label both ends W072; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R12 [CONVEYOR_IN4 pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W077 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.GND **Destination:** R17.2 **Identification:** Black; label both ends W077; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R17 [DRV_A.ENA local EN pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W078 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.GND **Destination:** R18.2 **Identification:** Black; label both ends W078; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R18 [DRV_A.ENB local EN pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W090 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.GND **Destination:** C_BA.- **Identification:** Black; label both ends W090; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_BA [local motor-bus reservoir; C / ESR / voltage TBD]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W095 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.GND **Destination:** Z_A.- **Identification:** Black; label both ends W095; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W188 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.OUT1 **Destination:** FM5.IN **Identification:** Orange; label both ends W188; DRV_A_OUT1 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM5 [motor 5 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W189 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.OUT1 **Destination:** FM6.IN **Identification:** Orange; label both ends W189; DRV_A_OUT1 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM6 [motor 6 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W190 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.OUT2 **Destination:** M5.B **Identification:** Orange; label both ends W190; DRV_A_OUT2 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W191 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.OUT2 **Destination:** M6.B **Identification:** Orange; label both ends W191; DRV_A_OUT2 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W192 | EXTERNAL CONNECTION - gate dependent **Source:** FM5.OUT **Destination:** M5.A **Identification:** Orange; label both ends W192; M5_A **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM5 [motor 5 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FM5; value/time curve TBD.	**Required electrical rating:** Bridge output after fuse; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W193 | EXTERNAL CONNECTION - gate dependent **Source:** FM6.OUT **Destination:** M6.A **Identification:** Orange; label both ends W193; M6_A **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM6 [motor 6 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FM6; value/time curve TBD.	**Required electrical rating:** Bridge output after fuse; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W194 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.OUT3 **Destination:** FM7.IN **Identification:** Orange; label both ends W194; DRV_A_OUT3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM7 [motor 7 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W195 | EXTERNAL CONNECTION - gate dependent **Source:** DRV_A.OUT4 **Destination:** M7.B **Identification:** Orange; label both ends W195; DRV_A_OUT4 **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: DC motor-lead fuse and rated branch/driver; recirculation diodes required.	**Required electrical rating:** Bidirectional / PWM bridge output; not a supply rail; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify motor current, polarity/direction, pigtail rating and overload coordination; no motor trial yet. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W196 | EXTERNAL CONNECTION - gate dependent **Source:** FM7.OUT **Destination:** M7.A **Identification:** Orange; label both ends W196; M7_A **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FM7 [motor 7 fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: FM7; value/time curve TBD.	**Required electrical rating:** Bridge output after fuse; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: One motor lead per fuse; no breadboard path. Motor voltage/direction still unverified. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.

7. Three HC-SR04 sensors

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

Label US_F, US_L and US_R. Photograph each module's actual VCC/TRIG/ECHO/GND order. Front uses GPIO32/34, left 33/35 and right 13/36 for TRIG/ECHO respectively.

Each VCC uses qualified protected logic 5V and each GND the logic return. TRIG comes through U1. Every ECHO uses its own 2.2k upper/3.3k lower divider and then U2; no direct 5V-to-GPIO connection is permitted.

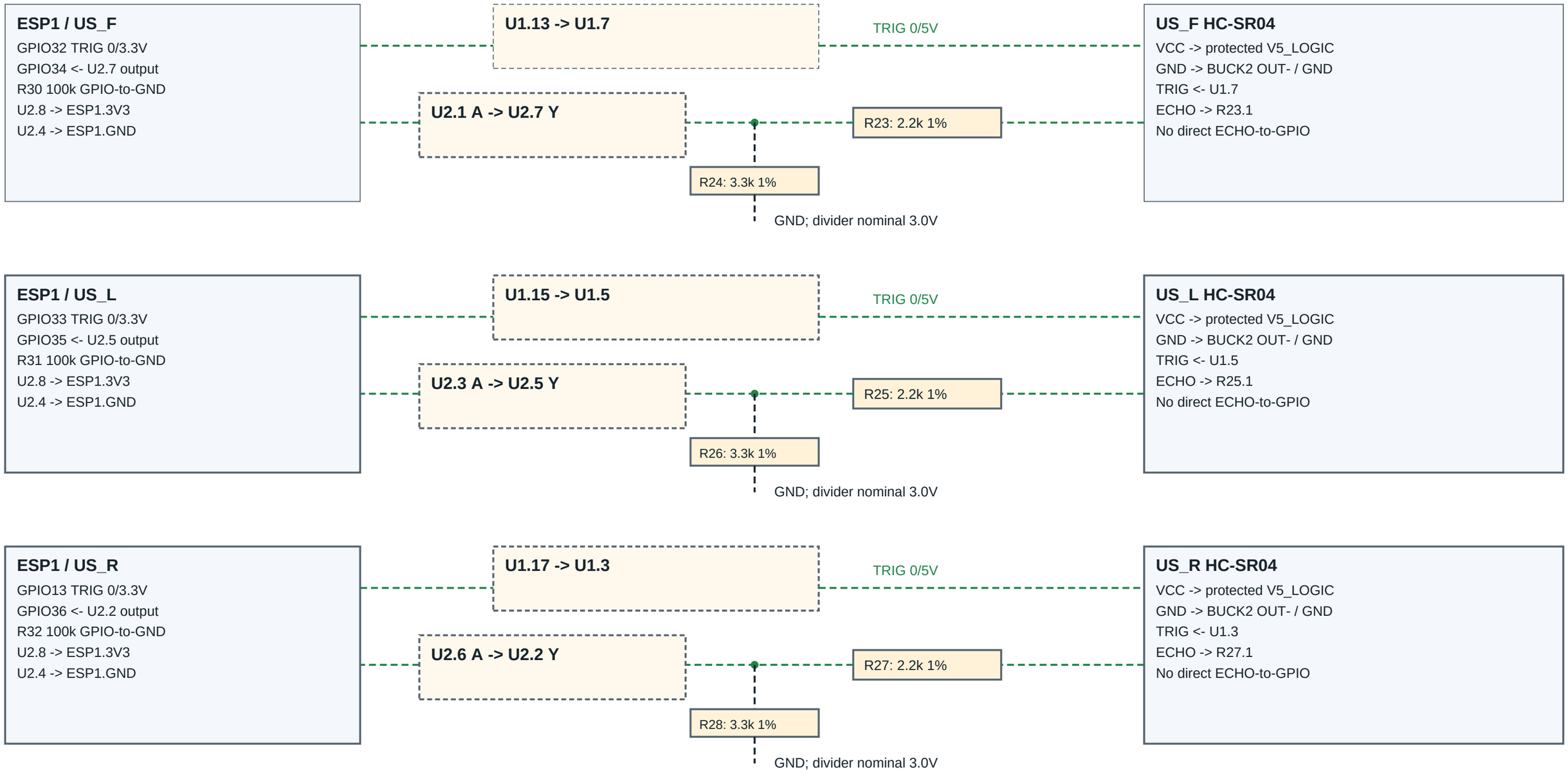
Fit the dividers and local 100nF capacitors on the detached interface assembly. Leave all actual ECHO plugs to the MCU disconnected until raw/divided/buffered waveform, edge-rate and partial-power tests pass.

After electrical qualification, test all three sensors together for fresh valid ranges and acoustic crosstalk. Wokwi's ideal direct ECHO wires and LED indicators are not a physical wiring reference.

HC-SR04: dividers plus powered-off ECHO isolation

U2 = SN74LVC3G34DCU; supply from MCU 3V3, not sensor 5V. Same GPIOs. Verify actual pulse levels/edges before connection.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026



C_U2: 100nF U2.8-to-U2.4. U2 inputs: HIGH >=2.0V, LOW <=0.8V; edge rate <=10ns/V at 3.3V +/-0.3V. Keep divider-to-U2 traces short. Scope actual sensor cable and pulse shape. U2 outputs follow MCU supply; power-off Ioff limits backfeed.

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Detailed connection records - Three HC-SR04 sensors

A record is one E2 segment. Endpoint resistor/fuse/contact bodies continue into other records; do not install duplicate components or jumpers. Color identifies function; voltage and ratings are requirements, not measurements.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W028 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.OUT+ **Destination:** US_F.VCC **Identification:** Red; label both ends W028; V5_LOGIC **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; qualify 4.75-5.25 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W029 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.OUT+ **Destination:** US_L.VCC **Identification:** Red; label both ends W029; V5_LOGIC **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; qualify 4.75-5.25 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W030 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.OUT+ **Destination:** US_R.VCC **Identification:** Red; label both ends W030; V5_LOGIC **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; qualify 4.75-5.25 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W032 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.OUT+ **Destination:** C_F.1 **Identification:** Red; label both ends W032; V5_LOGIC **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_F [local ceramic bypass; 100nF, >=10V]; OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; qualify 4.75-5.25 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W033 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.OUT+ **Destination:** C_L.1 **Identification:** Red; label both ends W033; V5_LOGIC **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_L [local ceramic bypass; 100nF, >=10V]; OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; qualify 4.75-5.25 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W034 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.OUT+ **Destination:** C_R.1 **Identification:** Red; label both ends W034; V5_LOGIC **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_R [local ceramic bypass; 100nF, >=10V]; OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; qualify 4.75-5.25 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W050 | EXTERNAL CONNECTION - gate dependent **Source:** BUCK2.OUT- **Destination:** US_F.GND **Identification:** Black; label both ends W050; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W051 | EXTERNAL CONNECTION - gate dependent **Source:** BUCK2.OUT- **Destination:** US_L.GND **Identification:** Black; label both ends W051; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W052 | EXTERNAL CONNECTION - gate dependent **Source:** BUCK2.OUT- **Destination:** US_R.GND **Identification:** Black; label both ends W052; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W085 | EXTERNAL CONNECTION - gate dependent **Source:** US_F.GND **Destination:** C_F.2 **Identification:** Black; label both ends W085; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_F [local ceramic bypass; 100nF, >=10V]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W086 | EXTERNAL CONNECTION - gate dependent **Source:** US_L.GND **Destination:** C_L.2 **Identification:** Black; label both ends W086; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_L [local ceramic bypass; 100nF, >=10V]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W087 | EXTERNAL CONNECTION - gate dependent **Source:** US_R.GND **Destination:** C_R.2 **Identification:** Black; label both ends W087; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_R [local ceramic bypass; 100nF, >=10V]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W134 | EXTERNAL CONNECTION - gate dependent **Source:** U1.7 (2Y2) **Destination:** US_F.TRIG **Identification:** Green; label both ends W134; TF_5V **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W137 | EXTERNAL CONNECTION - gate dependent **Source:** U1.5 (2Y3) **Destination:** US_L.TRIG **Identification:** Green; label both ends W137; TL_5V **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W140 | EXTERNAL CONNECTION - gate dependent **Source:** U1.3 (2Y4) **Destination:** US_R.TRIG **Identification:** Green; label both ends W140; TR_5V **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W161 | EXTERNAL CONNECTION - gate dependent **Source:** US_F.ECHO **Destination:** R23.1 **Identification:** Green; label both ends W161; EF_RAW **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R23 [ECHO upper resistor; 2.2k 1%, >=1/8 W]. Surrounding protection: 2.2k upper / 3.3k lower 1% divider required.	**Required electrical rating:** Standard module: potentially 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Keep actual ECHO disconnected until parts fitted and high/low/overshoot verified independently of ESP32. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W162 | EXTERNAL CONNECTION - gate dependent **Source:** R23.2 **Destination:** R24.1 **Identification:** Green; label both ends W162; EF_DIV **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R23 [ECHO upper resistor; 2.2k 1%, >=1/8 W]; R24 [ECHO lower resistor; 3.3k 1%, >=1/8 W]. Surrounding protection: 2.2k/3.3k 1% divider then U2 power-off isolation.	**Required electrical rating:** Nominal 3.0 V at 5.0V ECHO; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W163 | EXTERNAL CONNECTION - gate dependent **Source:** R23.2 **Destination:** U2.1 (1A) **Identification:** Green; label both ends W163; EF_DIV **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R23 [ECHO upper resistor; 2.2k 1%, >=1/8 W]. Surrounding protection: 2.2k/3.3k 1% divider then U2 power-off isolation.	**Required electrical rating:** Nominal 3.0 V at 5.0V ECHO; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W166 | EXTERNAL CONNECTION - gate dependent **Source:** US_L.ECHO **Destination:** R25.1 **Identification:** Green; label both ends W166; EL_RAW **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R25 [ECHO upper resistor; 2.2k 1%, >=1/8 W]. Surrounding protection: 2.2k upper / 3.3k lower 1% divider required.	**Required electrical rating:** Standard module: potentially 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Keep actual ECHO disconnected until parts fitted and high/low/overshoot verified independently of ESP32. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W167 | EXTERNAL CONNECTION - gate dependent **Source:** R25.2 **Destination:** R26.1 **Identification:** Green; label both ends W167; EL_DIV **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R25 [ECHO upper resistor; 2.2k 1%, >=1/8 W]; R26 [ECHO lower resistor; 3.3k 1%, >=1/8 W]. Surrounding protection: 2.2k/3.3k 1% divider then U2 power-off isolation.	**Required electrical rating:** Nominal 3.0 V at 5.0V ECHO; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W168 | EXTERNAL CONNECTION - gate dependent **Source:** R25.2 **Destination:** U2.3 (2A) **Identification:** Green; label both ends W168; EL_DIV **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R25 [ECHO upper resistor; 2.2k 1%, >=1/8 W]. Surrounding protection: 2.2k/3.3k 1% divider then U2 power-off isolation.	**Required electrical rating:** Nominal 3.0 V at 5.0V ECHO; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W171 | EXTERNAL CONNECTION - gate dependent **Source:** US_R.ECHO **Destination:** R27.1 **Identification:** Green; label both ends W171; ER_RAW **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R27 [ECHO upper resistor; 2.2k 1%, >=1/8 W]. Surrounding protection: 2.2k upper / 3.3k lower 1% divider required.	**Required electrical rating:** Standard module: potentially 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Keep actual ECHO disconnected until parts fitted and high/low/overshoot verified independently of ESP32. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W172 | EXTERNAL CONNECTION - gate dependent **Source:** R27.2 **Destination:** R28.1 **Identification:** Green; label both ends W172; ER_DIV **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R27 [ECHO upper resistor; 2.2k 1%, >=1/8 W]; R28 [ECHO lower resistor; 3.3k 1%, >=1/8 W]. Surrounding protection: 2.2k/3.3k 1% divider then U2 power-off isolation.	**Required electrical rating:** Nominal 3.0 V at 5.0V ECHO; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W173 | EXTERNAL CONNECTION - gate dependent **Source:** R27.2 **Destination:** U2.6 (3A) **Identification:** Green; label both ends W173; ER_DIV **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R27 [ECHO upper resistor; 2.2k 1%, >=1/8 W]. Surrounding protection: 2.2k/3.3k 1% divider then U2 power-off isolation.	**Required electrical rating:** Nominal 3.0 V at 5.0V ECHO; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: At U2 input: HIGH >=2.0V, LOW <=0.8V, 0..5.5V and edge rate <=10ns/V. Measure actual sensor VOH/VOL/overshoot.. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

8. Logic buffer and ECHO protection

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

U1 is SN74AHCT244N, not HC: VCC20 to protected logic5V, GND10 and OE1/OE19 to ground. Preserve the exact pin map shown. R5 is specifically 2.2k at GPIO14/U1.11; other input/default-low resistors follow E2.

U2 is SN74LVC3G34DCU on a verified VSSOP8 carrier: pin8 ONLY to MCU3V3, pin4 to MCU ground, channels 1->7 front, 3->5 left, 6->2 right. A carrier header number is not an IC pin number; continuity-map every header pad before installing the chip or harness.

C_U1/C_U2 are local 100nF bypasses; R30-R32 are 100k U2-output-to-ground pulls. Six separate 10k driver-enable pulls remain at their receivers. Fit each item by reference rather than counting generic resistors.

Unpowered work may include a detached protected interface board, inspection, resistor checks and endpoint labeling after actual package orientation and carrier mapping are known. Keep it disconnected from the ESP32, sensors, drivers and every supply until the applicable checks pass.

Later measurements must establish U2 input HIGH >=2.0V, LOW <=0.8V and <=10ns/V input transitions at the specified 3.3V supply range; verify output levels and absence of unsafe partial-power/backfeed behavior. Do not substitute a generic shifter.

Buffers and default-low interface

Equal net names join. Exact connections and protection: assembly_connections sheets / READY_TO_WIRE.md. Ratings remain HOLD.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

U1 | SN74AHCT244N 3.3-to-5 V buffer

ADDITIONAL / MISSING OR SELECT

SN74AHCT244N

1 / 1OE_N	GND
2 / 1A1	L_FWD_3V3
3 / 2Y4	TR_5V
4 / 1A2	L_REV_3V3
5 / 2Y3	TL_5V
6 / 1A3	R_FWD_3V3
7 / 2Y2	TF_5V
8 / 1A4	R_REV_3V3
9 / 2Y1	ENABLE_5V
10 / GND	GND
11 / 2A1	ENABLE_3V3
12 / 1Y4	R_REV_5V
13 / 2A2	TF_3V3
14 / 1Y3	R_FWD_5V
15 / 2A3	TL_3V3
16 / 1Y2	L_REV_5V
17 / 2A4	TR_3V3
18 / 1Y1	L_FWD_5V
19 / 2OE_N	GND
20 / VCC	V5_LOGIC

R18 | DRV_A.ENB local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R32 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	ER_SAFE
2	GND

U2 | SN74LVC3G34DCU ECHO isolation

ADDITIONAL / MISSING OR SELECT

SN74LVC3G34DCU

1 / 1A	EF_DIV
2 / 3Y	ER_SAFE
3 / 2A	EL_DIV
4 / GND	GND
5 / 2Y	EL_SAFE
6 / 3A	ER_DIV
7 / 1Y	EF_SAFE
8 / VCC	V3V3

R14 | DRV_L.L_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R16 | DRV_R.L_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R30 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	EF_SAFE
2	GND

R5 | ENABLE input pull-down

ADDITIONAL / MISSING OR SELECT

2.2k 1%, >=1/8 W

1	ENABLE_3V3
2	GND

R13 | DRV_L.R_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R15 | DRV_R.R_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R17 | DRV_A.ENA local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R31 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	EL_SAFE
2	GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Detailed connection records - Logic buffer and ECHO protection

A record is one E2 segment. Endpoint resistor/fuse/contact bodies continue into other records; do not install duplicate components or jumpers. Color identifies function; voltage and ratings are requirements, not measurements.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W024 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.OUT+ **Destination:** U1.20 (VCC) **Identification:** Red; label both ends W024; V5_LOGIC **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; qualify 4.75-5.25 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W031 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.OUT+ **Destination:** C_U1.1 **Identification:** Red; label both ends W031; V5_LOGIC **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_U1 [local ceramic bypass; 100nF, >=10V]; OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Independent OVP/OCP/supervision; qualify voltage envelope and fault response.	**Required electrical rating:** Regulated 5.0 V target; qualify 4.75-5.25 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No MCU-connected 5V signal; verify ripple/load response and all actual board logic supply requirements. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W053 | EXTERNAL CONNECTION - gate dependent **Source:** BUCK2.OUT- **Destination:** U1.10 (GND) **Identification:** Black; label both ends W053; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W054 | EXTERNAL CONNECTION - gate dependent **Source:** U1.10 (GND) **Destination:** U1.1 (1OE_N) **Identification:** Black; label both ends W054; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W055 | EXTERNAL CONNECTION - gate dependent **Source:** U1.10 (GND) **Destination:** U1.19 (2OE_N) **Identification:** Black; label both ends W055; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W061 | EXTERNAL CONNECTION - gate dependent **Source:** U1.10 (GND) **Destination:** R1.2 **Identification:** Black; label both ends W061; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R1 [L_FWD input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W062 | EXTERNAL CONNECTION - gate dependent **Source:** U1.10 (GND) **Destination:** R2.2 **Identification:** Black; label both ends W062; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R2 [L_REV input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W063 | EXTERNAL CONNECTION - gate dependent **Source:** U1.10 (GND) **Destination:** R3.2 **Identification:** Black; label both ends W063; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R3 [R_FWD input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W064 | EXTERNAL CONNECTION - gate dependent **Source:** U1.10 (GND) **Destination:** R4.2 **Identification:** Black; label both ends W064; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R4 [R_REV input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W065 | EXTERNAL CONNECTION - gate dependent **Source:** U1.10 (GND) **Destination:** R5.2 **Identification:** Black; label both ends W065; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R5 [ENABLE input pull-down; 2.2k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W066 | EXTERNAL CONNECTION - gate dependent **Source:** U1.10 (GND) **Destination:** R6.2 **Identification:** Black; label both ends W066; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R6 [TF input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W067 | EXTERNAL CONNECTION - gate dependent **Source:** U1.10 (GND) **Destination:** R7.2 **Identification:** Black; label both ends W067; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R7 [TL input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W068 | EXTERNAL CONNECTION - gate dependent **Source:** U1.10 (GND) **Destination:** R8.2 **Identification:** Black; label both ends W068; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R8 [TR input pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W084 | EXTERNAL CONNECTION - gate dependent **Source:** U1.10 (GND) **Destination:** C_U1.2 **Identification:** Black; label both ends W084; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_U1 [local ceramic bypass; 100nF, >=10V]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W097 | EXTERNAL CONNECTION - gate dependent **Source:** U2.4 (GND) **Destination:** C_U2.2 **Identification:** Black; label both ends W097; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): C_U2 [local ceramic bypass; 100nF, >=10V]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W116 | EXTERNAL CONNECTION - gate dependent **Source:** U1.18 (1Y1) **Destination:** R19.1 **Identification:** Blue; label both ends W116; L_FWD_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R19 [L_FWD receiver pull-down; 10k 1%, >=1/8 W]. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W120 | EXTERNAL CONNECTION - gate dependent **Source:** U1.16 (1Y2) **Destination:** R20.1 **Identification:** Blue; label both ends W120; L_REV_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R20 [L_REV receiver pull-down; 10k 1%, >=1/8 W]. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W124 | EXTERNAL CONNECTION - gate dependent **Source:** U1.14 (1Y3) **Destination:** R21.1 **Identification:** Blue; label both ends W124; R_FWD_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R21 [R_FWD receiver pull-down; 10k 1%, >=1/8 W]. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W128 | EXTERNAL CONNECTION - gate dependent **Source:** U1.12 (1Y4) **Destination:** R22.1 **Identification:** Blue; label both ends W128; R_REV_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R22 [R_REV receiver pull-down; 10k 1%, >=1/8 W]. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W165 | EXTERNAL CONNECTION - gate dependent **Source:** U2.7 (1Y) **Destination:** R30.1 **Identification:** Green; label both ends W165; EF_SAFE **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R30 [ECHO output default-low; 100k 1%, >=1/8 W]. Surrounding protection: U2 Ioff supplied by ESP1.3V3; 100k output bias.	**Required electrical rating:** 0 / MCU-derived 3.3V; U2 off -> high impedance; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W170 | EXTERNAL CONNECTION - gate dependent **Source:** U2.5 (2Y) **Destination:** R31.1 **Identification:** Green; label both ends W170; EL_SAFE **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R31 [ECHO output default-low; 100k 1%, >=1/8 W]. Surrounding protection: U2 Ioff supplied by ESP1.3V3; 100k output bias.	**Required electrical rating:** 0 / MCU-derived 3.3V; U2 off -> high impedance; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W175 | EXTERNAL CONNECTION - gate dependent **Source:** U2.2 (3Y) **Destination:** R32.1 **Identification:** Green; label both ends W175; ER_SAFE **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R32 [ECHO output default-low; 100k 1%, >=1/8 W]. Surrounding protection: U2 Ioff supplied by ESP1.3V3; 100k output bias.	**Required electrical rating:** 0 / MCU-derived 3.3V; U2 off -> high impedance; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify U2 VCC equals ESP32 VDD; output HIGH >=0.75*VDD, LOW <=0.25*VDD; no phantom MCU power with sensor rail live. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

9. Buck converters

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

Label BUCK1 MCU and BUCK2 logic; record actual IN+/IN-/OUT+/OUT- markings and module components. W042/W044 are existing negative-bond inspection records, not new jumpers.

Both input branches F1/F2 originate on the protected battery distribution BEFORE K1. They must remain powered when the independent motor contactor opens. Never parallel their positive outputs.

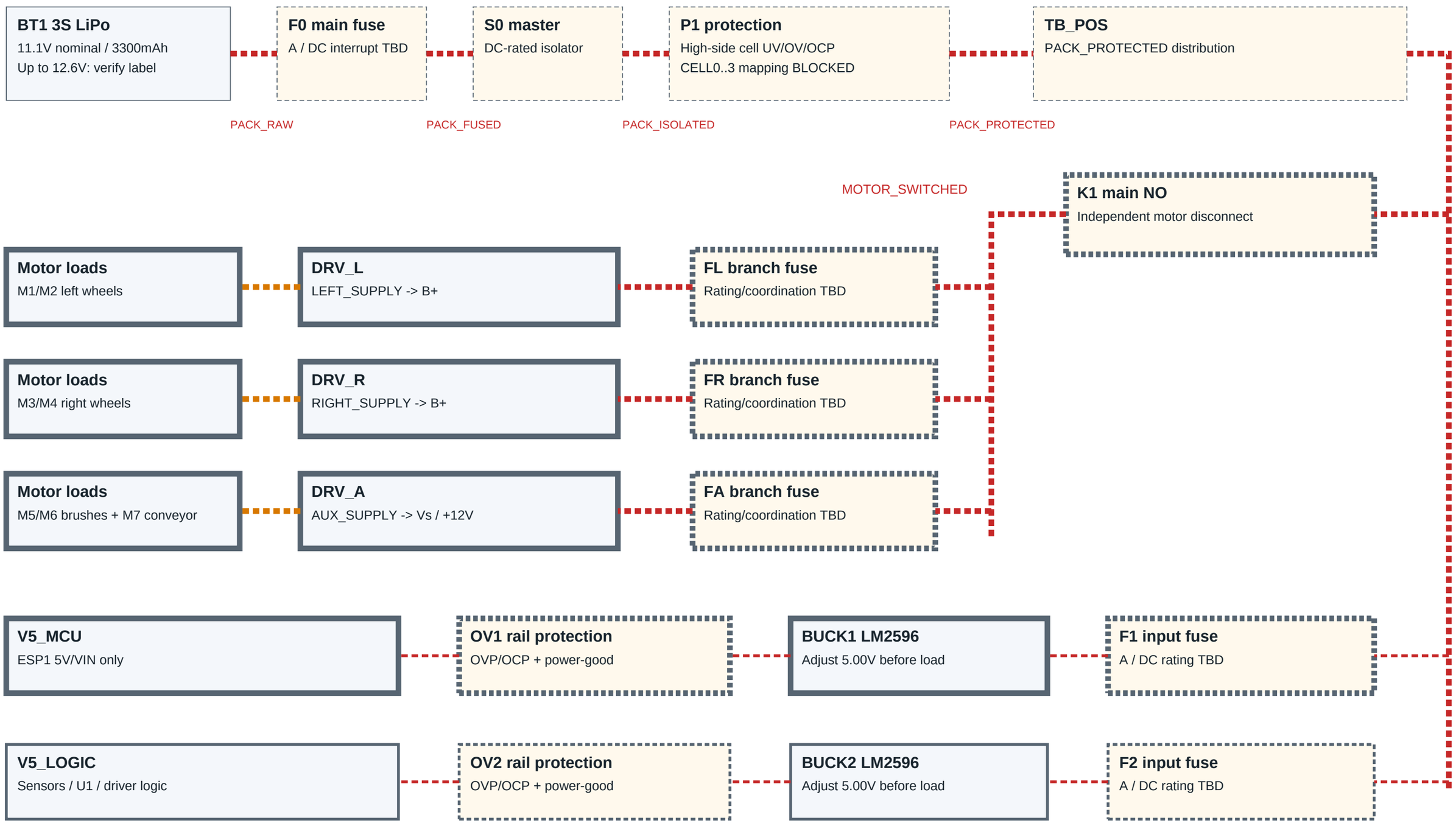
BUCK1 output goes through OV1 to the verified DevKit 5V/VIN input; BUCK2 through OV2 to logic/sensors. OV1/OV2 are required protection/supervision assemblies awaiting real part selection and pin maps; never replace them with wire links.

Later, use an approved current-limited bench source with all loads detached. Adjust each to 5.00V, qualify polarity/load/ripple/temperature/OVP and power-good behavior before attaching electronics. The LM2596 chip headline does not qualify either carrier.

Battery and power distribution

High-current paths are thick red/black. All fuse, wire, coil, protection and motor-current ratings are TBD; no battery operation is approved.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026



All returns -> TB_GND. Heavy driver returns; logic returns via buck OUT-. FK/coil chain: emergency_stop sheet. Do not parallel +5V outputs.

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Detailed connection records - Buck converters

A record is one E2 segment. Endpoint resistor/fuse/contact bodies continue into other records; do not install duplicate components or jumpers. Color identifies function; voltage and ratings are requirements, not measurements.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W006 | EXTERNAL CONNECTION - gate dependent **Source:** TB_POS.BUS **Destination:** F1.IN **Identification:** Red; label both ends W006; PACK_PROTECTED **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): F1 [MCU buck input fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: Cell-level protection + F0 + downstream fuses.	**Required electrical rating:** Loaded pack voltage; up to 12.6 V; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Select P1 and current ratings; do not use undefined raw BMS terminals. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W007 | EXTERNAL CONNECTION - gate dependent **Source:** TB_POS.BUS **Destination:** F2.IN **Identification:** Red; label both ends W007; PACK_PROTECTED **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): F2 [logic buck input fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: Cell-level protection + F0 + downstream fuses.	**Required electrical rating:** Loaded pack voltage; up to 12.6 V; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Select P1 and current ratings; do not use undefined raw BMS terminals. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W021 | EXTERNAL CONNECTION - gate dependent **Source:** F1.OUT **Destination:** BUCK1.IN+ **Identification:** Red; label both ends W021; MCU_BUCK_IN **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): F1 [MCU buck input fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: F1 for buck input and lead.	**Required electrical rating:** Pack voltage; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify module input range and fault-current/wire coordination. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W022 | EXTERNAL CONNECTION - gate dependent **Source:** F2.OUT **Destination:** BUCK2.IN+ **Identification:** Red; label both ends W022; LOGIC_BUCK_IN **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): F2 [logic buck input fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: F2 for buck input and lead.	**Required electrical rating:** Pack voltage; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify buck input range and module load budget. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W041 | EXTERNAL CONNECTION - gate dependent **Source:** TB_GND.BUS **Destination:** BUCK1.IN- **Identification:** Black; label both ends W041; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W043 | EXTERNAL CONNECTION - gate dependent **Source:** TB_GND.BUS **Destination:** BUCK2.IN- **Identification:** Black; label both ends W043; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W091 | EXTERNAL CONNECTION - gate dependent **Source:** BUCK1.OUT- **Destination:** OV1.GND **Identification:** Black; label both ends W091; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV1 [5V rail 1 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W092 | EXTERNAL CONNECTION - gate dependent **Source:** BUCK2.OUT- **Destination:** OV2.GND **Identification:** Black; label both ends W092; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W197 | EXTERNAL CONNECTION - gate dependent **Source:** BUCK1.OUT+ **Destination:** OV1.IN+ **Identification:** Red; label both ends W197; V5_MCU_RAW **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV1 [5V rail 1 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: OV1 required; never assume LM2596 is an overvoltage protector.	**Required electrical rating:** 5V target; buck-fault voltage may approach pack; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Select actual independent output protection and supervisor; simulate abnormal voltage with electronics disconnected. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W198 | EXTERNAL CONNECTION - gate dependent **Source:** BUCK2.OUT+ **Destination:** OV2.IN+ **Identification:** Red; label both ends W198; V5_LOGIC_RAW **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: OV2 required; never assume LM2596 is an overvoltage protector.	**Required electrical rating:** 5V target; buck-fault voltage may approach pack; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Select actual independent output protection and supervisor; simulate abnormal voltage with electronics disconnected. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

10. Battery protection and distribution

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

Prepare mounting space, wire-ID labels and cable-length records now. Keep the battery main and balance connectors unplugged and insulated; do not terminate a battery or balance harness to an unidentified protection board.

E2 topology is BT1 -> F0 adjacent main fuse -> S0 -> high-side cell protection P1 -> TB_POS. Pre-K1 branches feed both bucks and the coil chain; K1's motor bus feeds FL/FR/FA. Return currents use the rated ground star and dedicated heavy branches.

P1 CELL0..3 are electrical potentials, not a balance-connector order. Identify chemistry, cell count, charge limits, discharge capability, source fault envelope and the protection manufacturer's sense wiring before selecting or wiring this assembly.

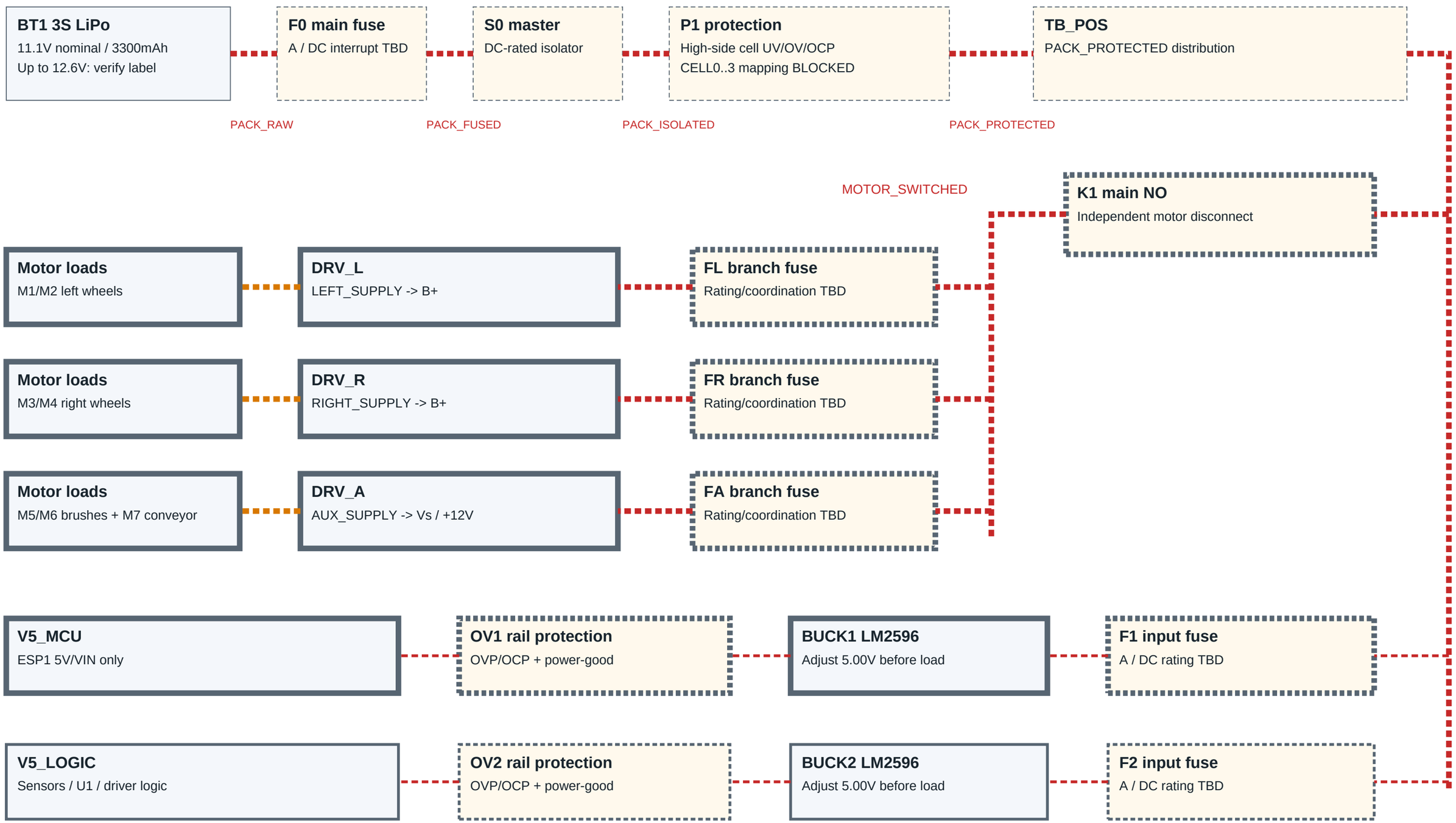
All fuse currents, conductor gauges, connectors, contact ratings and energy-clamp/reservoir selections remain conditional. Measure harness lengths and installation conditions without cutting motor pigtails. Numeric selection must coordinate actual load/start/fault clearing and smallest conductor/connector capacity.

Expected voltages in the connection register are design expectations, not measured values. The pack maximum of about 12.6V is conditional on ordinary 3S 4.2V-per-cell chemistry; the actual pack is not identified.

Battery and power distribution

High-current paths are thick red/black. All fuse, wire, coil, protection and motor-current ratings are TBD; no battery operation is approved.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026



All returns -> TB_GND. Heavy driver returns; logic returns via buck OUT-. FK/coil chain: emergency_stop sheet. Do not parallel +5V outputs.

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Detailed connection records - Battery protection and distribution

A record is one E2 segment. Endpoint resistor/fuse/contact bodies continue into other records; do not install duplicate components or jumpers. Color identifies function; voltage and ratings are requirements, not measurements.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W001 | EXTERNAL CONNECTION - gate dependent **Source:** BT1.PACK+ **Destination:** F0.IN **Identification:** Red; label both ends W001; PACK_RAW **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): F0 [main pack fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: Main fuse adjacent to pack; insulated connector.	**Required electrical rating:** Pack: up to 12.6 V; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify chemistry, polarity, pack C rating and connector interrupt/short-circuit capability. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W002 | EXTERNAL CONNECTION - gate dependent **Source:** F0.OUT **Destination:** S0.COM **Identification:** Red; label both ends W002; PACK_FUSED **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): F0 [main pack fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: F0 protects feed.	**Required electrical rating:** Pack voltage; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Select F0 from measured/spec start/stall currents and conductor ampacity. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W003 | EXTERNAL CONNECTION - gate dependent **Source:** S0.OUT **Destination:** P1.IN+ **Identification:** Red; label both ends W003; PACK_ISOLATED **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): P1 [3S high-side protection assembly; SELECT PART]. Surrounding protection: Master isolation plus P1.	**Required electrical rating:** Pack voltage; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify selected protection assembly input/output polarity and high-side topology. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W004 | EXTERNAL CONNECTION - gate dependent **Source:** P1.OUT+ **Destination:** TB_POS.BUS **Identification:** Red; label both ends W004; PACK_PROTECTED **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): P1 [3S high-side protection assembly; SELECT PART]. Surrounding protection: Cell-level protection + F0 + downstream fuses.	**Required electrical rating:** Loaded pack voltage; up to 12.6 V; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Select P1 and current ratings; do not use undefined raw BMS terminals. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W038 | EXTERNAL CONNECTION - gate dependent **Source:** TB_GND.BUS **Destination:** BT1.PACK- **Identification:** Black; label both ends W038; GND **Intermediate components:** No unnamed inline part in this segment. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W039 | EXTERNAL CONNECTION - gate dependent **Source:** TB_GND.BUS **Destination:** P1.GND **Identification:** Black; label both ends W039; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): P1 [3S high-side protection assembly; SELECT PART]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W101 | EXTERNAL CONNECTION - gate dependent **Source:** BT1.BAL_CELL0 **Destination:** P1.CELL0 **Identification:** Green; label both ends W101; CELL0 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): P1 [3S high-side protection assembly; SELECT PART]. Surrounding protection: Sense lead protection and connector order per selected P1.	**Required electrical rating:** Cell-stack potential; order TBD; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W102 | EXTERNAL CONNECTION - gate dependent **Source:** BT1.BAL_CELL1 **Destination:** P1.CELL1 **Identification:** Green; label both ends W102; CELL1 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): P1 [3S high-side protection assembly; SELECT PART]. Surrounding protection: Sense lead protection and connector order per selected P1.	**Required electrical rating:** Cell-stack potential; order TBD; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W103 | EXTERNAL CONNECTION - gate dependent **Source:** BT1.BAL_CELL2 **Destination:** P1.CELL2 **Identification:** Green; label both ends W103; CELL2 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): P1 [3S high-side protection assembly; SELECT PART]. Surrounding protection: Sense lead protection and connector order per selected P1.	**Required electrical rating:** Cell-stack potential; order TBD; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W104 | EXTERNAL CONNECTION - gate dependent **Source:** BT1.BAL_CELL3 **Destination:** P1.CELL3 **Identification:** Green; label both ends W104; CELL3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): P1 [3S high-side protection assembly; SELECT PART]. Surrounding protection: Sense lead protection and connector order per selected P1.	**Required electrical rating:** Cell-stack potential; order TBD; SIGNAL / LOGIC / COIL - size TBD. Signal/regulated branch only; actual load, leakage/pull network and conductor/connector suitability must be established. No numeric gauge/current invented. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: No connector pin order assumed. Verify cumulative cell potentials without probing adjacent contacts with bare tips. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.

11. Emergency-stop wiring

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

Select actual DC-rated K1 main NO and two independent NO auxiliaries, suitable coil, S0, latching S2 with three isolated NC functions, S3 momentary NO, suppression and rail-healthy interfaces before wiring their functional ports.

Coil route: FK -> OV1 healthy contact -> OV2 healthy contact -> S2 coil NC -> (S3 NO parallel K1 hold NO) -> K1 coil -> ground. Main K1 NO cuts all three motor branches independently of ESP32 software.

The separate S2 enable NC interrupts U1.9 to all six driver enables. The separate S2 sense NC and K1 sense NO carry MCU3V3 to GPIO27 with its pull-down. LOW is unsafe; no pull-up substitution or shared coil/sense contact is acceptable.

Serial ESTOP/STOP disables firmware outputs but does not open K1 or isolate motor power. With K1 open at boot, GPIO27 LOW intentionally latches ESTOP. Physical release requires verified disabled EN, deliberate hardware power-arm, healthy feedback, RESET, ARM and then explicit motion.

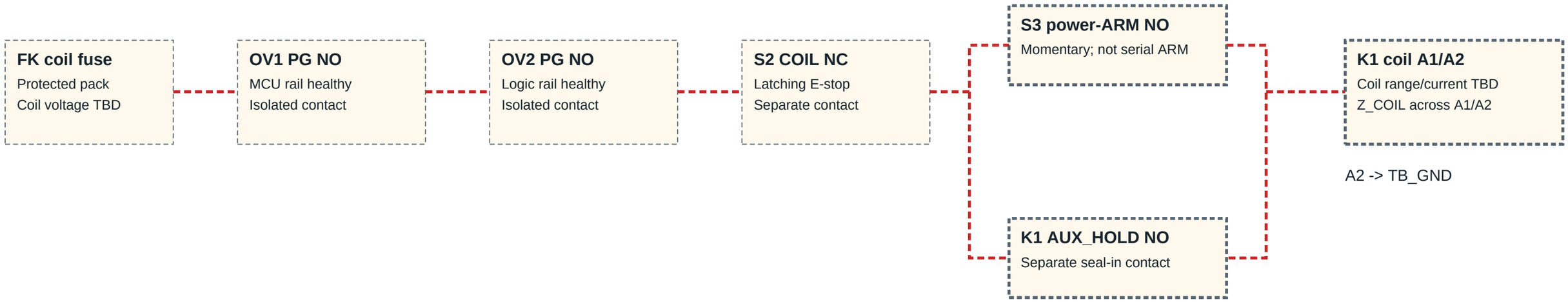
A held/stuck S3 or welded contacts are outside this simple seal-in circuit's fault coverage. Keep the existing E2 HOLD until actual components, risk assumptions, dropout, stored energy and restart tests have evidence. No safety certification is implied.

Independent emergency cutoff and manual power-arm

Functional contact roles only; actual contactor/supervisor/contact-block parts and terminal numbers must be selected and verified.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

1. Coil seal-in: independent of ESP32; S3 must be released for normal no-auto-restart behavior.



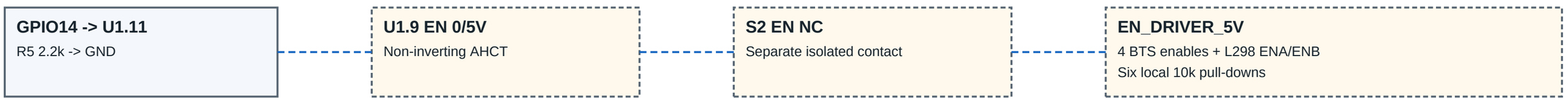
K1 MAIN_NO powers FL/FR/FA only. Stop opens coil + EN; K1 drops its seal-in and isolated feedback. Release alone does not restore power with S3 released. Held S3 / welded contacts are not covered by this functional single-channel circuit.

2. Firmware healthy input: HIGH only when E-stop released AND K1 closed.



K1-open boot intentionally latches ESTOP. Later: EN measured LOW -> deliberate power-ARM -> GPIO27 HIGH -> RESET -> ARM.

3. Shared enable: physical S2 contact opens every driver EN, independently of software.



Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Detailed connection records - Emergency-stop wiring

A record is one E2 segment. Endpoint resistor/fuse/contact bodies continue into other records; do not install duplicate components or jumpers. Color identifies function; voltage and ratings are requirements, not measurements.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W005 | EXTERNAL CONNECTION - gate dependent **Source:** TB_POS.BUS **Destination:** K1.MAIN_COM **Identification:** Red; label both ends W005; PACK_PROTECTED **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): K1 [DC motor contactor; CONTACT / COIL TBD]. Surrounding protection: Cell-level protection + F0 + downstream fuses.	**Required electrical rating:** Loaded pack voltage; up to 12.6 V; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Select P1 and current ratings; do not use undefined raw BMS terminals. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W008 | EXTERNAL CONNECTION - gate dependent **Source:** TB_POS.BUS **Destination:** FK.IN **Identification:** Red; label both ends W008; PACK_PROTECTED **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FK [contactor coil fuse / holder; DC A / interrupt rating TBD]. Surrounding protection: Cell-level protection + F0 + downstream fuses.	**Required electrical rating:** Loaded pack voltage; up to 12.6 V; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Select P1 and current ratings; do not use undefined raw BMS terminals. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W009 | EXTERNAL CONNECTION - gate dependent **Source:** K1.MAIN_NO **Destination:** FL.IN **Identification:** Red; label both ends W009; MOTOR_SWITCHED **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FL [left driver fuse / holder; DC A / interrupt rating TBD]; K1 [DC motor contactor; CONTACT / COIL TBD]. Surrounding protection: Independent motor power removal.	**Required electrical rating:** Pack voltage when K1 closed; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Prove E-stop drops K1, release alone does not re-latch, and stored bus energy decays. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W010 | EXTERNAL CONNECTION - gate dependent **Source:** K1.MAIN_NO **Destination:** FR.IN **Identification:** Red; label both ends W010; MOTOR_SWITCHED **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FR [right driver fuse / holder; DC A / interrupt rating TBD]; K1 [DC motor contactor; CONTACT / COIL TBD]. Surrounding protection: Independent motor power removal.	**Required electrical rating:** Pack voltage when K1 closed; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Prove E-stop drops K1, release alone does not re-latch, and stored bus energy decays. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W011 | EXTERNAL CONNECTION - gate dependent **Source:** K1.MAIN_NO **Destination:** FA.IN **Identification:** Red; label both ends W011; MOTOR_SWITCHED **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FA [auxiliary driver fuse / holder; DC A / interrupt rating TBD]; K1 [DC motor contactor; CONTACT / COIL TBD]. Surrounding protection: Independent motor power removal.	**Required electrical rating:** Pack voltage when K1 closed; HIGH CURRENT - size TBD. Current/AWG/mm2 NOT SELECTED: use actual run/start/stall, protective clearing, weakest pigtail/connector, route length/drop, ambient/bundling and thermal limits. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Prove E-stop drops K1, release alone does not re-latch, and stored bus energy decays. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W035 | EXTERNAL CONNECTION - gate dependent **Source:** ESP1.3V3 **Destination:** S2.SENSE_COM **Identification:** Red; label both ends W035; V3V3 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: MCU-derived 3V3; dry feedback and U2 only; never V5_LOGIC.	**Required electrical rating:** ESP32 3.3 V; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify no connection to 5V/coil circuit and actual feedback-contact minimum current. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W056 | EXTERNAL CONNECTION - gate dependent **Source:** TB_GND.BUS **Destination:** K1.A2 **Identification:** Black; label both ends W056; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): K1 [DC motor contactor; CONTACT / COIL TBD]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W057 | EXTERNAL CONNECTION - gate dependent **Source:** K1.A2 **Destination:** Z_COIL.A2 **Identification:** Black; label both ends W057; GND **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): K1 [DC motor contactor; CONTACT / COIL TBD]. Surrounding protection: Star power returns, isolated signal routing; never switch only GND.	**Required electrical rating:** 0 V reference; returns sized by branch; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W105 | EXTERNAL CONNECTION - gate dependent **Source:** FK.OUT **Destination:** OV1.PG_COM **Identification:** Red; label both ends W105; COIL_FUSED **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): FK [contactor coil fuse / holder; DC A / interrupt rating TBD]; OV1 [5V rail 1 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: FK + S2 coil NC.	**Required electrical rating:** Pack voltage; K1 coil compatibility TBD; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Coil must operate across actual loaded pack range, or a qualified coil supply is required. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_RATING; actual evidence PENDING.
Record W106 | EXTERNAL CONNECTION - gate dependent **Source:** S2.COIL_NC **Destination:** S3.COM **Identification:** Red; label both ends W106; COIL_STOP_CHAIN **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]; S3 [Momentary motor-power ARM; NO; rating TBD]. Surrounding protection: Independent NC E-stop chain.	**Required electrical rating:** Coil supply when stop released; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Pressing stop must break coil circuit even if MCU hung. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W107 | EXTERNAL CONNECTION - gate dependent **Source:** S2.COIL_NC **Destination:** K1.AUX_HOLD_COM **Identification:** Red; label both ends W107; COIL_STOP_CHAIN **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): K1 [DC motor contactor; CONTACT / COIL TBD]; S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Independent NC E-stop chain.	**Required electrical rating:** Coil supply when stop released; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Pressing stop must break coil circuit even if MCU hung. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W108 | EXTERNAL CONNECTION - gate dependent **Source:** S3.NO **Destination:** K1.AUX_HOLD_NO **Identification:** Red; label both ends W108; COIL_HOLD **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): K1 [DC motor contactor; CONTACT / COIL TBD]; S3 [Momentary motor-power ARM; NO; rating TBD]. Surrounding protection: Momentary S3 in parallel with K1 NO auxiliary.	**Required electrical rating:** Coil supply when manually latched; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify seal-in, dropout, no automatic restart on stop release/power restoration, and clamp release time. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W109 | EXTERNAL CONNECTION - gate dependent **Source:** S3.NO **Destination:** K1.A1 **Identification:** Red; label both ends W109; COIL_HOLD **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): K1 [DC motor contactor; CONTACT / COIL TBD]; S3 [Momentary motor-power ARM; NO; rating TBD]. Surrounding protection: Momentary S3 in parallel with K1 NO auxiliary.	**Required electrical rating:** Coil supply when manually latched; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify seal-in, dropout, no automatic restart on stop release/power restoration, and clamp release time. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W110 | EXTERNAL CONNECTION - gate dependent **Source:** S3.NO **Destination:** Z_COIL.A1 **Identification:** Red; label both ends W110; COIL_HOLD **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): S3 [Momentary motor-power ARM; NO; rating TBD]. Surrounding protection: Momentary S3 in parallel with K1 NO auxiliary.	**Required electrical rating:** Coil supply when manually latched; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify seal-in, dropout, no automatic restart on stop release/power restoration, and clamp release time. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W111 | EXTERNAL CONNECTION - gate dependent **Source:** K1.AUX_SENSE_NO **Destination:** ESP1.GPIO27 **Identification:** Blue; label both ends W111; ESTOP_OK **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): K1 [DC motor contactor; CONTACT / COIL TBD]. Surrounding protection: Isolated NC contact, R29 pull-down; firmware internal pull-down.	**Required electrical rating:** Healthy 3.3 V; open/pressed 0 V; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: HIGH only when S2 released AND K1 closed. Open contact/wire or power-good loss trips; initial K1-open boot latches ESTOP until manual power-arm, RESET, ARM.. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W112 | EXTERNAL CONNECTION - gate dependent **Source:** K1.AUX_SENSE_NO **Destination:** R29.1 **Identification:** Blue; label both ends W112; ESTOP_OK **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): K1 [DC motor contactor; CONTACT / COIL TBD]; R29 [E-stop feedback pull-down; 10k 1%, >=1/8 W]. Surrounding protection: Isolated NC contact, R29 pull-down; firmware internal pull-down.	**Required electrical rating:** Healthy 3.3 V; open/pressed 0 V; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: HIGH only when S2 released AND K1 closed. Open contact/wire or power-good loss trips; initial K1-open boot latches ESTOP until manual power-arm, RESET, ARM.. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W131 | EXTERNAL CONNECTION - gate dependent **Source:** U1.9 (2Y1) **Destination:** S2.EN_COM **Identification:** Blue; label both ends W131; ENABLE_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: 5V non-inverting AHCT output.	**Required electrical rating:** 0/5 V; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify target board accepts 5V logic; scope high/low margins and no opposing PWM. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W141 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** DRV_L.R_EN **Identification:** Blue; label both ends W141; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W142 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** DRV_L.L_EN **Identification:** Blue; label both ends W142; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W143 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** DRV_R.R_EN **Identification:** Blue; label both ends W143; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W144 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** DRV_R.L_EN **Identification:** Blue; label both ends W144; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W145 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** DRV_A.ENA **Identification:** Blue; label both ends W145; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W146 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** DRV_A.ENB **Identification:** Blue; label both ends W146; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W147 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** R13.1 **Identification:** Blue; label both ends W147; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R13 [DRV_L.R_EN local EN pull-down; 10k 1%, >=1/8 W]; S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W148 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** R14.1 **Identification:** Blue; label both ends W148; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R14 [DRV_L.L_EN local EN pull-down; 10k 1%, >=1/8 W]; S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W149 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** R15.1 **Identification:** Blue; label both ends W149; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R15 [DRV_R.R_EN local EN pull-down; 10k 1%, >=1/8 W]; S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W150 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** R16.1 **Identification:** Blue; label both ends W150; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R16 [DRV_R.L_EN local EN pull-down; 10k 1%, >=1/8 W]; S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W151 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** R17.1 **Identification:** Blue; label both ends W151; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R17 [DRV_A.ENA local EN pull-down; 10k 1%, >=1/8 W]; S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W152 | EXTERNAL CONNECTION - gate dependent **Source:** S2.EN_NC **Destination:** R18.1 **Identification:** Blue; label both ends W152; EN_DRIVER_5V **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): R18 [DRV_A.ENB local EN pull-down; 10k 1%, >=1/8 W]; S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Six local 10k pull-downs; EN jumpers removed; independent S2 NC.	**Required electrical rating:** 0/5 V; LOW disabled; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Calculate total input/pull load <= U1 drive and receiver VIH/VIL; no hard-HIGH module links; scope reset sequence. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W199 | EXTERNAL CONNECTION - gate dependent **Source:** OV1.PG_NO **Destination:** OV2.PG_COM **Identification:** Red; label both ends W199; COIL_PG1 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV1 [5V rail 1 OVP/OCP and supervision assembly; SELECT / VERIFY]; OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]. Surrounding protection: Both logic rails must be healthy before K1 can energize/hold.	**Required electrical rating:** Coil supply; contact isolation required; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: A missing/abnormal rail opens the coil chain; restoration alone must not re-latch K1. Select actual supervisor/interface circuitry.. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W200 | EXTERNAL CONNECTION - gate dependent **Source:** OV2.PG_NO **Destination:** S2.COIL_COM **Identification:** Red; label both ends W200; COIL_PG2 **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): OV2 [5V rail 2 OVP/OCP and supervision assembly; SELECT / VERIFY]; S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Both logic rails must be healthy before K1 can energize/hold.	**Required electrical rating:** Coil supply; contact isolation required; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: A missing/abnormal rail opens the coil chain; restoration alone must not re-latch K1. Select actual supervisor/interface circuitry.. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.
Record W201 | EXTERNAL CONNECTION - gate dependent **Source:** S2.SENSE_NC **Destination:** K1.AUX_SENSE_COM **Identification:** Blue; label both ends W201; SENSE_STOP_OK **Intermediate components:** No unnamed inline part in this segment. Endpoint component(s): K1 [DC motor contactor; CONTACT / COIL TBD]; S2 [Latching E-stop / 3 isolated NC contacts; SELECT PART]. Surrounding protection: Separate dry K1 NO feedback contact in series with S2 NC.	**Required electrical rating:** 3.3 V only; SIGNAL / LOGIC / COIL - size TBD. Coil/contact or isolated control circuit: actual current, low-current wetting, contact isolation and coil range TBD per selected part. Never mix coil and GPIO contacts. Insulation/connector voltage must cover actual maximum normal and fault/transient exposure; expected voltage is not measured evidence. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Map actual contact terminals; keep isolated from seal-in coil circuit. No jumper around this interlock.. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** BLOCKED_MISSING; actual evidence PENDING.

12. Final component purchase list

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

The fixed selections and quantities below come from E2's component model and FINAL_PURCHASE_LIST.md. Buy those identified logic/passive items now; their physical installation still needs package/orientation inspection.

The VSSOP8 adapter/carrier can be specified from U2's known package, but purchase an actual model only after its mechanical drawing, IC-pad map and header map are checked. Labels and insulation supplies may be obtained as ordinary assembly supplies; do not treat them as qualification of power wiring.

The complete conditional purchase table is reproduced from the authoritative E2 list. Do not buy fuses, wire, contactor, clamps, high-current replacement drivers or a BMS from guessed currents or a photograph alone.

L298 replacement is conditional, not selected: E2 identifies two Pololu G2 18v17 #2991 as candidates after load/thermal/protection qualification. A replacement requires a documented design revision, wiring/model/diagrams and adapter/Wokwi semantics checks, not an undocumented swap.

Purchase now: fixed E2 selections

Component	Qty	Reference labels	Mounting / specification note
SN74AHCT244N	1	U1	Exact 20-pin pinout per TI. No PCB footprint assigned; package must match procured part. This is AHCT, not HC.
SN74LVC3G34DCU	1	U2	TI SCES366L 8-pin VSSOP pin map. 5.5V tolerant inputs and Ioff; supply ONLY from ESP1 3V3. Fit verified adapter. Input edges must meet <=10ns/V at VCC=3.3+/-0.3V; locate beside dividers and qualify waveform. Not a generic bidirectional MOSFET shifter.
2.2k 1%, >=1/8 W resistor	4	R23, R25, R27, R5	1% tolerance, at least 1/8W, documented actual value/package.
3.3k 1%, >=1/8 W resistor	3	R24, R26, R28	1% tolerance, at least 1/8W, documented actual value/package.
10k 1%, >=1/8 W resistor	22	R1, R2, R3, R4, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R29	1% tolerance, at least 1/8W, documented actual value/package.
100nF, >=10V ceramic capacitor	5	C_U1, C_F, C_L, C_R, C_U2	Fit actual package/mounting; effective capacitance at bias must meet E2.
100k 1%, >=1/8 W resistor	3	R30, R31, R32	1% tolerance, at least 1/8W, documented actual value/package.

This is 2 ICs, 32 resistors and 5 capacitors; adapter purchase requires verified package-to-header mapping. Motor/power hardware remains conditional.

Complete E2 conditional purchase specification

Component / references	Quantity	Required characteristics and purpose	Verified proposed part / alternatives	Mandatory?	Evidence dependency
3.3-to-5V buffer U1	1	Non-inverting octal, TTL-compatible 2.0V HIGH / 0.8V LOW inputs; 4.5-5.5V supply; 20-pin DIP for practical assembly	TI **SN74AHCT244N**; same function AHCT package only with verified adapter; HC is unsuitable	Yes	Datasheet pinout fixed; actual receiver loading/defaults remain to verify
ECHO isolation U2	1	Triple non-inverting buffer; 5.5V-tolerant input, powered-off Ioff; supply from MCU 3V3	TI **SN74LVC3G34DCU**, 8-pin VSSOP; DCT variant needs a separately mapped adapter	Yes	Verified pin map; qualify actual sensor edges <=10ns/V and partial-power behavior
VSSOP8 adapter / carrier	1	Match U2 DCU mechanical drawing, short divider traces, local decoupling; insulated support	Select actual adapter after package/header map review; no footprint assumed	Yes	Header-to-IC continuity and geometry
10k bias resistors R1-4, R6-22, R29	22	1%, >=1/8W; defined LOW at U1 inputs, drivers, GPIO27	Any documented resistor meeting value/tolerance/power and chosen mounting	Yes	Module pull-ups must not defeat them; six EN pulls load U1
2.2k resistors R5, R23, R25, R27	4	1%, >=1/8W; R5 dominates GPIO14 reset bias; three ECHO series uppers	Documented 2.2k parts	Yes	GPIO14 waveform and sensor VOH/overshoot
3.3k resistors R24, R26, R28	3	1%, >=1/8W; ECHO node-to-GND lower resistors	Documented 3.3k parts	Yes	Fit each complete divider; never series resistor alone
100k resistors R30-32	3	1%, >=1/8W; U2 output/GPIO bias; ~33uA load at 3.3V	Documented 100k parts	Yes	Total output load including GPIO leakage <100uA for cited VOH bound
Ceramic bypass C_U1, C_U2, C_F, C_L, C_R	5	100nF, >=10V, close to named receiver pins; dielectric/capacitance valid at bias	Documented ceramic parts, package selected for mounting	Yes	Verify existing module bypassing; these five remain explicit external placements
Driver reservoir capacitors C_BL/C_BR/C_BA	3 positions	Capacitance, ESR, ripple-current and working voltage from transient/regen qualification	**Unselected**; no guessed uF/voltage/footprint	Yes, unless verified onboard provision satisfies the revised design	Rail peaks, bus impedance, braking energy, ambient
Motor-bus clamps Z_L/Z_R/Z_A	3 positions	Standoff above normal pack maximum, worst clamp below each module limit; pulse/thermal energy capacity; correctly coordinated fuse	**Unselected** TVS / qualified energy-absorbing network	Yes, or explicitly reviewed equivalent	Regenerated energy, all module limits and fuse behavior; TVS alone may be insufficient
Main/branch/coil fuses and DC holders F0, FL, FR, FA, F1, F2, FK	7 each	DC voltage and interrupt capacity covering actual source; time-current/I2t coordination with smallest protected conductor and load	**Unselected** currents/series/holders	Yes	Pack fault capability, start currents, conductor/connector ratings
Individual motor fuses and holders FM1-7	7 each	DC and bipolar bridge-output service; coordinate lead/motor overload limits and branch protection	**Unselected**; 14 total fuse positions in current model	Yes under current topology	Each motor's pigtail, run/start/stall duration and clearing curves; not semiconductor protection by assumption
Master isolator S0	1	DC load-break capability at worst normal/fault envelope; insulated mounting	**Unselected**	Yes	Total current, source voltage, interruption and inrush
3S protection P1	1 complete assembly	Cell-specific UV/OV, overcurrent/short protection, high-side disconnection; protected sense leads and documented cell-tap map; proper discharge/regen behavior	**Unselected**; documented protected 3S replacement pack is an alternative requiring a model revision	Yes	Actual battery chemistry, per-cell limits, continuous C rating, total demand; no unidentified low-side BMS

Component / references	Quantity	Required characteristics and purpose	Verified proposed part / alternatives	Mandatory?	Evidence dependency
Rail protectors OV1/OV2	2 complete assemblies	Independent output OVP/OCP and undervoltage/power-good; healthy regulated envelope 4.75-5.25V or tighter actual load limits; isolated fail-open coil-permissive contacts	**Unselected**; likely multiple components/PCB, not an assumed plug-in module	Yes	MCU/sensor peak budget; trip tolerances/delay; maximum buck fail-through; contact/coil load; rail ramp and fault behavior
Motor contactor K1	1 assembly	Main NO DC motor contact plus **two independent NO auxiliaries**, suitable coil over complete loaded 3S range; current/inrush/break ratings and dropout time	**Unselected**; actual main/aux terminal IDs required	Yes	Measured simultaneous current, braking/inrush, coil voltage/current range and release performance
Latching E-stop S2 with isolated contact blocks	1 operator + 3 NC contacts	Mechanically latching, deliberate release, coil-rated NC plus two low-current NC circuits; verify minimum wetting current at 3.3V	**Unselected**; actual contact block compatibility required	Yes	Coil suppression/current; sense and logic low-current reliability
Momentary hardware power-arm S3	1	NO pushbutton rated for coil current/inrush and environment	**Unselected**	Yes	K1 coil; basic seal-in does not reject a held/stuck S3
Coil suppression Z_COIL	1 network	Per K1 manufacturer; correct polarity/clamp energy; release time remains adequate	**Unselected** TVS/diode-zener network; plain diode only if dropout qualification permits	Yes	Coil inductance/current, allowed clamp voltage, stop time
Rated positive and ground distribution TB_POS/TB_GND	2 assemblies	Enough separately secured positions, barriers/covers; ground star with dedicated heavy returns	**Unselected**; multiple terminal blocks allowed with reviewed bridge ratings	Yes	Total/branch currents, conductor size, mechanical installation
Harness wire, connectors, lugs, ferrules, heatshrink and strain relief	1 measured harness set	Current, temperature, length and voltage-drop ratings; reverse-polarity prevention; no motor-current Dupont/breadboard paths	**Unselected** gauge/connector series	Yes	201 connection records (197 external + 4 internal checks), lengths, protection curves and pigtails
Mounting, guards, heatsinks and insulation	1 chassis set	Secure battery, guarded brushes/conveyor, insulated distribution; cooling sized for simultaneous operation	**Unselected**	Yes	Mechanical dimensions, thermal test and stopping-distance assessment
LiPo balance charger and safe charging arrangement	1, if absent	Exact pack chemistry/cell-count and maker-approved charge limits; correct balance connector	**Unselected**	Yes for operation	Battery label/datasheet/condition; never charge through an unspecified rover protection port
Current-limited bench source, DMM, scope, current and temperature instruments	Access to 1 suitable set	Voltage/current capacity and bandwidth appropriate to staged tests	Existing lab access acceptable	Yes for commissioning	Load/fault envelope; do not use battery for first measurements
Replacement cleaning drivers	0 initially; **2 if L298 fails**	Two independent documented MOSFET H bridges; continuous/start/stall/regen/thermal ratings covering brush-pair and conveyor separately; default-safe hardware disable	**Candidate: 2 x Pololu G2 18v17, #2991**; not approved until measured	Conditional mandatory	See replacement path below; a driver must not be selected from RPM alone
USB/UART data interface	1, if needed	Board-approved USB isolation arrangement, or qualified 3.3V UART with VCC disconnected and suitable grounding	**Unselected**; data-only cable is not automatic backfeed protection	Conditional mandatory	Actual DevKit USB/VIN schematic and power sequencing

L298 conditional replacement and required design revision

L298 suitability is HOLD, not proven FAIL. One channel sees both brush currents summed; the other sees conveyor current. Obtain load/start/stall profiles at actual approved supply voltage, diode ratings, heatsink thermal path and simultaneous case temperature. The 2A/channel DC figure is an absolute maximum; a board's reliable continuous allowance may be substantially lower. At the datasheet's worst 2A bridge drop of 4.9V, loss can reach 9.8W in one channel. Calculate both channels together and the motor's delivered voltage. Lower software PWM does not qualify an overload. ST L298 datasheet.

If the L298 fails, a practical **candidate** is two Pololu G2 18v17 #2991 boards. The manufacturer documents VIN, GND, OUTA, OUTB, PWM, DIR and SLP; its approximately 17A continuous claim depends on specified conditions and is not an approval for unknown loads. The default current-limit threshold is approximately 40A, so motor protection/VREF selection must be separately qualified. Manufacturer product and pinout.

Conditional revision only: GPIO21 -> brush PWM, GPIO22 -> brush DIR; GPIO23 -> conveyor PWM, GPIO25 -> conveyor DIR; gated EN -> both SLP inputs. Preserve GPIO14/27 and wheel/sensor pins. Remove/override each SLP pull-up according to the actual board revision with verified dominant default-low circuitry. VIN uses suitably split fused AUX supply; OUTA/OUTB replace OUT1/2 or OUT3/4; retain individual motor protection. Do not connect VM or 3V3(out) to external rails. Current/thermal and physical enable-disable qualification precede integration. Existing full-on forward-only auxiliary outputs could map HIGH PWM/LOW DIR, but adapter semantics, stop behavior, Wokwi labels, all schematics and regressions must be revised and reviewed before substitution. **This alternate topology is not the active E2 netlist and is not a ready-to-wire alternative.**

The U1 thresholds, pin map and package are from TI SN74AHCT244. U2 input thresholds at 3.0-3.6V, powered-off isolation, pin map, input-edge requirement and output limits are from TI SN74LVC3G34. These selections do not identify or approve unknown carrier boards. Battery capacity and nominal voltage do not establish chemistry or discharge capability; the actual pack documentation is still missing.

Missing measurements: collect data now; powered entries stay gated

ID / when	Item	Exact data needed	Acceptance / selection dependency
D01 / D now: labels/data	Actual modules and terminal maps	Board manufacturer/revision, every used printed terminal and orientation; U1/U2 package-to-adapter map; correct links/regulator mode; maker schematics for DevKit, BTS, L298 and bucks.	One-to-one E2 functional-port to actual printed terminal map, readable photos/datasheets. Unreadable/unmarked terminal remains unconnected.
D02 / D now: pack data	Battery specification and condition	Chemistry, cell count, charge/discharge limits, continuous/peak C and duration, condition, main/balance connector identification and manufacturer sense order; fault envelope from documented data, never a battery short test.	P1, charger, contact/fuse voltage/current and sense wiring selected against actual pack limits, not nominal 11.1V alone.
D03 / D data first; H later approved isolated tests	All seven motors	For each M1-M7: part/rated voltage and permitted range, load/run current, start peak and duration, stall current and permitted duration, insulation/pigtail limits; conveyor voltage specifically. Use manufacturer data or separately approved low-energy characterization, never hand-held stalling.	Voltage and RMS/run/start/stall envelopes fit motor, driver, connector and coordinated cutoff limits; missing current profile prevents final power selection.
D04 / D data / E selection; H later confirmation	Summed driver and thermal budgets	M1+M2, M3+M4 and M5+M6 totals, simultaneous M7 demand; documented board derating, delivered motor voltage, diode capacity, ambient/case/heatsink temperatures and thermal path; start/stall clearing limits.	All continuous/pulse/thermal/voltage limits met simultaneously with stated margin; L298 failing any limit triggers a reviewed replacement revision.
D05 / A/C/D unpowered; E selection	Harness, fuse and contact ratings	Each branch/pigtail material/size/connector rating, one-way route length, ambient/bundling, allowed voltage drop, start/fault I-squared-t and source DC interruption envelope; contactor main/inrush/break and coil voltage/current range across loaded pack.	Fuse clearing precedes smallest protected lead/motor/driver allowable overload; voltage drop/temperature acceptable; contacts and holders documented for DC service. No guessed AWG or fuse amperage.
D06 / C unpowered only	Continuity and isolation	Record test-lead resistance; each external segment end-to-end ohms, neighboring-net isolation and polarity with intended sensitive electronics detached; inspect four existing bonds separately. Identify any semiconductor path in context.	Expected path agrees with conductor/lead budget, no unintended cross-net link; lead-resistance-corrected values reviewed. Never megger connected electronics or meter resistance across a live battery.
D07 / G only after release	Regulators and rail protectors	Each buck output no-load/peak load, startup/turn-off peak, ripple, temperature and OVP/OCP/UV/power-good response over approved input range; raw and protected outputs separately, dummy loads first.	5.00V target; protected healthy envelope 4.75-5.25V or tighter actual part limits, rated peak load/heat, fail-open coil-permissive response verified. Scope fast peaks; do not intentionally expose MCU to a buck fault.
D08 / G only after release	ECHO, enable, reset and partial power	Raw/divided/U2-output HIGH/LOW, overshoot, rise/fall rate under actual harness, MCU/logic rail order and backfeed, GPIO14 and all six enables through boot/reset/brownout, summed receiver loads.	U2 input HIGH >=2.0V, LOW <=0.8V, 0-5.5V and <=10ns/V edges at specified VCC; GPIO levels within ESP32 limits; no unsafe EN pulse/backfeed or raw 5V GPIO path. Use isolated test fixtures before MCU attachment.
D09 / I only after release; repeat after H	Cutoff, restart and stored energy	K1 dropout/coil range, stop-contact continuity, enable fall, bus residual voltage/discharge, normal manual reset sequence, held-reset/weld limitations, suppression and worst braking/regeneration peak/energy.	Physical S2 removes all motor power independently of firmware, opens enable and drives GPIO27 LOW; release alone with S3 released does not restart. Clamp/reservoir and discharge/stopping limits supported by reviewed data/traces.
D10 / H/I/J only after release	Direction and actual vehicle operation	Each motor direction and permitted low-energy current, paired run/starting profiles and heat, brush inward sweep/conveyor transport, sensor/crosstalk ranges, actual speed/stopping distance at worst intended floor/load and supervision envelope.	Physical directions correct, currents/heat inside approved limits; no auto restart, safe distance margin and guards; controller/simulation results alone never satisfy these measurements.

13. Assembly verification checklist

Stage boundary: unpowered preparation permitted for identified items; physical power-up HOLD retained.

Start at checkpoint A. A and documented unpowered portions of B/C/D can proceed without the physical power-up HOLD being lifted. An unresolved terminal/rating stops that affected connection, not mounting, labeling or preparing other identified parts.

Recommended order: A -> B with component mapping -> C plus D -> E -> F -> G -> I motor-free cutoff test -> H -> I loaded confirmation -> J. Checkpoint letters organize the work; the independent cutoff must be proven before driver-connected motor testing.

All F-J energized stages are NOT RELEASED in this handover. No missing physical measurement is marked PASS. Supervisor release is per stage and records its evidence; a completed unpowered checklist or virtual test does not authorize battery or motor power.

Use TEAM_ASSEMBLY_RECORD.csv for every one of the 201 records, and TEAM_CHECKPOINT_RECORD.csv for stage sign-off. For each result record instrument/method, actual value and unit, approved limit, inspector/date and photo/trace path. BLANK/PENDING is not PASS.

Send a named evidence batch per checkpoint: overview, readable labels/terminal close-ups, wire-ID/probe endpoint photos for ambiguous or changed routes, complete CSV readings and original scope/thermal/current files for any later powered stage. A photo of a meter without probe endpoints and units cannot establish the measured net.

Checkpoint	Current permission	Result
A - Unpowered mounting and labeling	AVAILABLE: dry-fit and label, no electrical power	PENDING / not executed
B - Protected signal-interface assembly	UNPOWERED ONLY: after fixed parts and carrier map verified	PENDING / not executed
C - Unpowered wiring and continuity checks	AVAILABLE FOR IDENTIFIED SEGMENTS; unresolved routes stay open	PENDING / not executed
D - Physical component identification and measurements	LABELS/DATASHEETS AND UNPOWERED METHODS ONLY NOW	PENDING / not executed
E - Component-dependent electrical completion	HOLD: actual selection and mappings required; completion remains unpowered	PENDING / not executed
F - USB-only firmware installation	NOT RELEASED: future isolated DevKit procedure	PENDING / not executed
G - Regulated power testing	NOT RELEASED: staged current-limited bench tests only after evidence	PENDING / not executed
H - Individual motor testing	NOT RELEASED: approved low-energy tests; cutoff first for driver tests	PENDING / not executed
I - Emergency-stop testing	NOT RELEASED: motor-free preflight before H; loaded confirmation afterward	PENDING / not executed
J - Complete autonomous operation	NOT RELEASED: all earlier physical gates and supervised trial plan required	PENDING / not executed

Checkpoint A - Unpowered mounting and labeling

AVAILABLE: dry-fit and label, no electrical power

Prerequisites: All battery main/balance connectors, USB and bench sources disconnected; motor ends isolated. Use an unpowered chassis/plate; keep battery detached and protected.

1. Mark FRONT and label ESP1, DRV_L/R/A, BUCK1/2, US_F/L/R, M1-M7 and planned protection positions.

2. Dry-fit mounts and guards with clearance for moving parts; do not drill into/near a battery or unknown wiring. Photograph with a ruler and measure harness routes with string.

3. Print both-end wire-ID labels for the 197 external segments only. Mark the four internal-bond inspection IDs on the module worksheet, not on new wires. Preserve pigtails/colors; leave current-dependent circuits unterminated.

Evidence item	Required record
Photographs	P00 layout/underside/ruler; P07 motor mounts; overview showing detached battery and unplugged sources.
Measurements	Mount/hole/plate dimensions, guard and sensor clearance, one-way cable-route lengths; no powered electrical measurement in A.
PASS criteria	All actual items labeled, mechanical fit documented, separate routing possible, no powered connection; unknown parts have reserved space rather than guessed terminals.
STOP conditions	Any supply attached, damaged battery/pigtail, interference/pinch hazard, obscured labels or drilling near unknown cables; stop affected mounting and resolve.
Send to remote supervisor	A batch: P00/P07 images, reference inventory, layout with FRONT arrow, dimension/route-length sheet, list of missing items.
Actual result / acceptance	PENDING | assembler: _____ | reviewer: _____ | date: _____ | evidence batch: _____

Checkpoint B - Protected signal-interface assembly

UNPOWERED ONLY: after fixed parts and carrier map verified

Prerequisites: A done; correct U1/U2 and passive values available; actual package pin1 and adapter-to-IC mapping inspected. Interface remains detached from all modules and supplies.

1. Lay out U1 and U2 with local C_U1/C_U2; fit reference-labeled bias networks, R5 specifically 2.2k, three ECHO dividers and R30-R32 100k output pulls.

2. Keep divider-to-U2 nodes short. Label U2 pin8 MCU3V3 so it cannot be mistaken for logic5V; separate input/output connectors by documented pin maps.

3. Inspect top and underside for bridges, correct upper/lower divider routing and supply/ground polarity. Do not change E2 values or use a generic level shifter.

Evidence item	Required record
Photographs	P09 IC markings/pin1, carrier both sides, complete top/underside with reference labels; ruler/trace layout.
Measurements	Resistor values before installation or interpreted with parallel paths; adapter/header continuity, supply-to-ground isolation, divider and OE-to-ground mapping. No voltage applied.
PASS criteria	All fixed parts match E2, U1/U2 orientation and every adapter pad mapped; resistor/reference and supply routing verified with no unintended shorts.
STOP conditions	Unknown chip/package, unverified carrier map, wrong/missing resistor, raw ECHO bypass, U2 powered from logic5V in layout, solder bridge or unresolved low-resistance path.
Send to remote supervisor	B batch: P09 photos, carrier pin-map worksheet, passive-value/continuity readings, solder inspection and any deviation marked unresolved.
Actual result / acceptance	PENDING | assembler: _____ | reviewer: _____ | date: _____ | evidence batch: _____

Checkpoint C - Unpowered wiring and continuity checks

AVAILABLE FOR IDENTIFIED SEGMENTS; unresolved routes stay open

Prerequisites: A/B applicable checks done; actual endpoint labels mapped for each segment; conductor/connector requirements established for that segment. All sources and motor loads remain detached.

1. Build only identified, appropriately specified detached signal segments and mount/connect accepted passive interface parts. Label both ends with wire ID, net and terminal.

2. Leave battery/balance, all motor outputs and high-current branches, unselected safety ports, external buck/MCU power and all actual ECHO-to-MCU paths unconnected. Prepare their labels/lengths only.

3. Record each of 197 external segment checks without populating missing parts with links. Check W042/W044/W046/W048 as internal bonds; add no wire for them.

4. Use low-energy meter continuity/ohms with sensitive electronics detached; trace end-to-end and adjacent nets, not just a continuity buzzer. Preserve separately insulated NC pins.

Evidence item	Required record
Photographs	P12_C_overview and each terminal-strip/harness close-up; both end labels; probe endpoints/display for ambiguous or changed wire IDs.
Measurements	D06 for each completed segment; four internal-bond results separate. Record lead resistance, actual ohms, expected limit and interpretation; no megger on electronics.
PASS criteria	Completed subset agrees with register/netlist and verified maps, no cross-net shorts; open pending routes explicitly tagged and logged. Unpowered progress can PASS for that subset without releasing power.
STOP conditions	Unidentified pin, unset power-wire rating, missing protection, unexpected bond/short/polarity, untraceable splice, or any connected supply; stop that route, isolate and report.
Send to remote supervisor	C batch: TEAM_ASSEMBLY_RECORD.csv with completed/pending rows, terminal maps, harness photos and explicit unresolved wire-ID list.
Actual result / acceptance	PENDING | assembler: _____ | reviewer: _____ | date: _____ | evidence batch: _____

Checkpoint D - Physical component identification and measurements

LABELS/DATASHEETS AND UNPOWERED METHODS ONLY NOW

Prerequisites: Reference labels assigned; source isolation maintained. Powered current/ripple/thermal characterization is a later separately released test, not authorized by this checkpoint.

1. Collect P01-P11 images and manufacturer documents; fill functional-port-to-actual-terminal maps. Prioritize battery identity, conveyor voltage and all motor part numbers.

2. Collect D01-D06 data: ratings/current profiles from datasheets, harness geometry, permitted limits and unpowered continuity. Mark missing values PENDING rather than guessing from appearance or RPM.

3. If motor start/stall data is unavailable, obtain a supervised low-energy characterization plan with known voltage/current/duration limits before any such test. Never short-test the battery or hold a stalled gearbox by hand.

Evidence item	Required record
Photographs	All requested component identification views P01-P11, legible part/revision and actual terminal markings; no internal battery disassembly.
Measurements	Manufacturer motor run/start/stall and thermal limits; pack C/cell/chemistry data; cable/pigtail/connector geometry and unpowered bonds. D07-D10 remain future gated measurements.
PASS criteria	Each affected selection has traceable exact-part data or an explicitly accepted future measurement plan; missing data and affected routes clearly listed. Energized gates remain pending.
STOP conditions	Unreadable/unmatched board, unknown conveyor voltage, missing safe measurement limits, unexplained module links, damaged pack or temptation to infer current from a photo.
Send to remote supervisor	D batch: P01-P11 originals, manufacturer PDFs, terminal mapping, completed/remaining D01-D06 worksheet and current_budget.csv values with evidence.
Actual result / acceptance	PENDING | assembler: _____ | reviewer: _____ | date: _____ | evidence batch: _____

Checkpoint E - Component-dependent electrical completion

HOLD: actual selection and mappings required; completion remains unpowered

Prerequisites: D provides load/source/installation evidence; reviewed actual P1/OV1/OV2/K1/S0/S2/S3, fuse/wire/connector, clamp/reservoir and driver selections. Any replacement requires a recorded design revision first.

1. Select parts against approved load/start/stall/thermal/fault data and populate exact manufacturer terminal maps; no arbitrary fuse/contactor/TVS/BMS substitution.

2. Finish accepted harness branches while isolated, including heavy star returns, protection placements, independent coil/enable/sense contacts and verified module link settings. Keep battery connectors, actual motor loads and MCU/ECHO power harnesses detached for initial tests.

3. Repeat per-wire and adjacent-net checks; update the physical as-built record and have any topology/pin/value deviation explicitly reviewed against E2 before it is used.

Evidence item	Required record
Photographs	P10/P11 actual selections and markings, fuse/holder/terminal installation, every safety contact block, complete top/underside and revised terminal maps.
Measurements	Selection calculations D02-D05, unpowered continuity/insulation/contact truth states, maker torque/crimp requirements and allowed conductor sizes.
PASS criteria	No generic functional port remains guessed on completed circuits; ratings coordinated, driver arrangement qualified on data, complete as-built mapping and continuity accepted for the intended next test only.
STOP conditions	Unselected current-dependent part, missing protection or feedback contact, regulator/jumper conflict, overload/thermal failure, guessed balance order or undocumented design deviation.
Send to remote supervisor	E batch: approved BOM/rating calculations/datasheets, as-built terminal and connection records, protection/contact photos, explicit outstanding stage gates.
Actual result / acceptance	PENDING | assembler: _____ | reviewer: _____ | date: _____ | evidence batch: _____

Checkpoint F - USB-only firmware installation

NOT RELEASED: future isolated DevKit procedure

Prerequisites: D identifies actual DevKit/cable/port; supervisor has accepted a USB-only plan and source identity. Completely detach every external GPIO and power harness, battery/balance and motor wiring from the DevKit.

1. Build esp32dev from the reviewed E2 source and record commit/hash/target. Use a confirmed data-capable cable matching the actual connector; do not guess its type from this package.

2. Upload only to the isolated board under the separately released procedure; record upload log. Open monitor at 115200 baud with LF and verify disabled startup.

3. With GPIO27 open, ESTOP latching is expected. Do not jumper it HIGH or send movement commands to defeat the feedback design. Disconnect USB before reattaching any external harness.

Evidence item	Required record
Photographs	DevKit alone with every external connector detached; USB/board identification and monitor log capture.
Measurements	Recorded source/firmware identity, upload target/port and disabled startup/feedback status; any board voltage check only as allowed by the released isolated plan.
PASS criteria	Reviewed firmware installed on identified isolated board, startup disabled and no external backfeed path; reconnecting external hardware is not released by this PASS.
STOP conditions	Any external harness remains, unknown USB connector/board, upload mismatch, unexpected enabled startup, or attempt to bypass feedback.
Send to remote supervisor	F batch: detached-board photo, commit/build/upload logs, serial startup log and operator/date; preserve older installed firmware status until this is actually done.
Actual result / acceptance	PENDING | assembler: _____ | reviewer: _____ | date: _____ | evidence batch: _____

Checkpoint G - Regulated power testing

NOT RELEASED: staged current-limited bench tests only after evidence

Prerequisites: E accepted for intended test; F if MCU will be used; exact schematic, current limits, test points, dummy loads and supervisor release documented. Battery and motor loads remain disconnected; no mixed USB/external supply until qualified.

1. Test each detached buck and selected protector independently with approved isolated/current-limited source and dummy load. Set 5.00V and document polarity/load/ripple/temperature and protection behavior before attaching electronics.

2. Qualify U1/U2/sensor and enable interfaces on safe test fixtures, including reset/rail-order/backfeed/edge behavior, before connecting real GPIOs. Leave actual ECHO-to-MCU plugs open until interface acceptance.

3. Attach only the specifically accepted electronics at each substep. Record scope rails and all six enables; a rail issue or unreviewed power-order change cancels that substep.

Evidence item	Required record
Photographs	Source current limit and wiring, dummy load/test-point labels, each buck/protector, isolated interface fixture; scope probe points and settings.
Measurements	D07/D08: rail no-load/load/ripple/peaks/heat/protection, ECHO logic/edges, partial power and reset/default-low enable traces; original instrument files.
PASS criteria	All intended loads meet their documented rail/logic limits and protected supply response, no unsafe backfeed/enable glitch; accepted substeps recorded. Battery/motor tests remain separately gated.
STOP conditions	Wrong polarity, output outside accepted envelope, OVP/PG failure, unsafe EN pulse, raw 5V GPIO path, unexpected heat/backfeed or unknown current limit.
Send to remote supervisor	G batch: wiring/current-limit photos, measured table, original scope captures, ripple/load/thermal/protection records and accepted substep list.
Actual result / acceptance	PENDING | assembler: _____ | reviewer: _____ | date: _____ | evidence batch: _____

Checkpoint H - Individual motor testing

NOT RELEASED: approved low-energy tests; cutoff first for driver tests

Prerequisites: Known motor voltage and permitted current/duration/thermal limits; guarded restraints/lifted wheels and supervised plan. G and I motor-free cutoff test must PASS before any driver-connected motor trial. Battery remains disconnected unless separately released later.

1. Test one mechanically guarded motor at a time using approved current-limited bench energy, verified protection and current measurement setup. Do not insert a meter across a supply or hand-stall a gearbox.

2. Record direction at approved short pulses; isolate before changing A/B leads. Confirm all wheel forward arrows, inward brush sweep and conveyor-to-bin movement. Preserve reference and polarity records.

3. Only after individual envelopes pass, qualify summed wheel pairs and simultaneous cleaning demand under the reviewed plan; record start/run profiles, delivered voltage and temperatures. An unknown stall duration is a stop condition, not a test target.

Evidence item	Required record
Photographs	Each motor ID, restraints/guards, polarity and current-probe setup; direction video and mechanical path.
Measurements	D03/D04/D10 as released: actual voltage, start peak/duration, run/load current and thermal rise; compare with approved driver/motor/fuse/lead limits. Stall data by accepted method only.
PASS criteria	Directions and each measured operating envelope within approved limits; pair/cleaning qualification specifically documented before combined use. Any L298 failure leaves cleaning disconnected.
STOP conditions	No I motor-free PASS for driver-connected trials, unknown voltage/limits, jam, opposing wheel/brush direction, overcurrent/overheat, voltage sag or failed shutdown; isolate source.
Send to remote supervisor	H batch: per-motor/pair/cleaning tables, current/time/thermal originals, polarity updates, guarded direction videos and any required driver-revision proposal.
Actual result / acceptance	PENDING | assembler: _____ | reviewer: _____ | date: _____ | evidence batch: _____

Checkpoint I - Emergency-stop testing

NOT RELEASED: motor-free preflight before H; loaded confirmation afterward

Prerequisites: E/G accepted, actual contactor/contacts/suppression and limits documented, current-limited test source plus safe dummy load. All motors disconnected for first cutoff test. Loaded confirmation requires H acceptance.

1. First test the physical coil/enable/sense chain using a safe dummy load, independently of firmware. Prove K1-open boot, manual power-arm, S2 interruption and GPIO27 truth states.

2. Measure dropout, motor-bus discharge and all enable states; release S2 with S3 released and verify no automatic reseal. Verify serial ESTOP only disables firmware outputs and leaves K1 bus electrically distinct.

3. Before H, record motor-free cutoff acceptance. After H, repeat under the approved motor/load envelope for regeneration/stopping and stored energy; document held-S3/welded-contact limitations rather than claiming fault tolerance.

Evidence item	Required record
Photographs	Complete labeled contact chain, separate coil/enable/sense contacts, source/dummy load, scope points; later guarded stop video.
Measurements	D09: coil range, dropout/enable/feedback timing, residual voltage/discharge, normal reset sequence and clamping/braking peaks against approved limits.
PASS criteria	All motor branches removed independently of MCU, enables interrupted, GPIO27 LOW unsafe, no restart on S2 release alone with S3 released; measured response/energy within accepted limits. Record motor-free and loaded outcomes separately.
STOP conditions	Any powered branch survives S2 unexpectedly, false HIGH feedback, missing independent contact, automatic reseal, unsafe stored energy/clamp/dropout, or unsupported risk assumption.
Send to remote supervisor	I batch: contact maps, dummy-load preflight record before H, original timing/voltage traces, reset-sequence video and later loaded confirmation; no current result is prefilled PASS.
Actual result / acceptance	PENDING | assembler: _____ | reviewer: _____ | date: _____ | evidence batch: _____

Checkpoint J - Complete autonomous operation

NOT RELEASED: all earlier physical gates and supervised trial plan required

Prerequisites: A-I accepted for actual as-built configuration, H summed-load and I loaded results closed, installed firmware identity known, guards/controlled area/physical-stop operator present. Battery use requires its own accepted pack/protection/charger and power-up evidence.

1. Begin with the reviewed controlled first-power procedure and low-speed supervised trials, clear area and stop operator. Verify three fresh ranges and physical stop before movement.

2. Explicit RESET/ARM/AUTO only under the accepted plan; verify forward/turn/avoidance, bounded recovery, independent cleaning while forward, sensor faults, manual timeout and no restart after stop/reset.

3. Measure actual speed/stopping margin on intended floor/load. Keep operations supervised until the documented physical envelope and all acceptance limits have been demonstrated; record faults instead of bypassing interlocks.

Evidence item	Required record
Photographs	Complete as-built top/underside, all guards/covers, battery protection installation, test area and accessible physical stop; continuous trial video.
Measurements	D10 plus full-system rail/pack sag, current/thermal/regeneration and stopping-distance records under worst intended approved conditions; sensor freshness/crosstalk and stop/restart logs.
PASS criteria	All physical acceptance limits met, no unresolved safety/protection/thermal/wiring hold for this configuration, correct garbage collection directions and bounded autonomous behavior under the stated supervised envelope.
STOP conditions	Any previous gate pending, person/obstacle risk, stale sensor, runaway/restart, loose harness/guard, overcurrent/heat/rail fault or unexplained discrepancy; use physical stop and isolate.
Send to remote supervisor	J batch: final as-built photos/records, installed commit/hash, trial logs/video, quantitative speed/current/heat/stop data and signed configuration-specific verdict.
Actual result / acceptance	PENDING | assembler: _____ | reviewer: _____ | date: _____ | evidence batch: _____

Four internal-bond verification records - NO ADDITIONAL WIRES

A record is one E2 segment. Endpoint resistor/fuse/contact bodies continue into other records; do not install duplicate components or jumpers. Color identifies function; voltage and ratings are requirements, not measurements.

Connection / actual endpoints / intermediate protection	Required rating / verification / retained E2 dependency
Record W042 | INSPECT EXISTING BOND - DO NOT ADD WIRE **Source:** BUCK1.IN- **Destination:** BUCK1.OUT- **Identification:** Black is the E2 reference color; GND; existing bond only, no new wire or end labels to fit. **Intermediate components:** Existing internal module bond. NO WIRE TO ADD and no extra intermediate component. Inspect against the actual module schematic and unpowered reading.	**Required electrical rating:** N/A: do not purchase or size another conductor for this existing bond. Verify the actual module and any shared-current path; do not infer its capacity from continuity. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W044 | INSPECT EXISTING BOND - DO NOT ADD WIRE **Source:** BUCK2.IN- **Destination:** BUCK2.OUT- **Identification:** Black is the E2 reference color; GND; existing bond only, no new wire or end labels to fit. **Intermediate components:** Existing internal module bond. NO WIRE TO ADD and no extra intermediate component. Inspect against the actual module schematic and unpowered reading.	**Required electrical rating:** N/A: do not purchase or size another conductor for this existing bond. Verify the actual module and any shared-current path; do not infer its capacity from continuity. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W046 | INSPECT EXISTING BOND - DO NOT ADD WIRE **Source:** DRV_L.B- **Destination:** DRV_L.GND **Identification:** Black is the E2 reference color; GND; existing bond only, no new wire or end labels to fit. **Intermediate components:** Existing internal module bond. NO WIRE TO ADD and no extra intermediate component. Inspect against the actual module schematic and unpowered reading.	**Required electrical rating:** N/A: do not purchase or size another conductor for this existing bond. Verify the actual module and any shared-current path; do not infer its capacity from continuity. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.
Record W048 | INSPECT EXISTING BOND - DO NOT ADD WIRE **Source:** DRV_R.B- **Destination:** DRV_R.GND **Identification:** Black is the E2 reference color; GND; existing bond only, no new wire or end labels to fit. **Intermediate components:** Existing internal module bond. NO WIRE TO ADD and no extra intermediate component. Inspect against the actual module schematic and unpowered reading.	**Required electrical rating:** N/A: do not purchase or size another conductor for this existing bond. Verify the actual module and any shared-current path; do not infer its capacity from continuity. **Verification:** UNPOWERED FIRST: isolate all sources and relevant loads; trace marked endpoints, record ohms/polarity and adjacent-net checks. E2 acceptance: Verify actual buck common negatives; no motor return via ESP32/USB/breadboard. Any scope/voltage/load test is FUTURE and stage-gated, not authorized by this row. **E2 status retained:** PROVISIONAL; actual evidence PENDING.